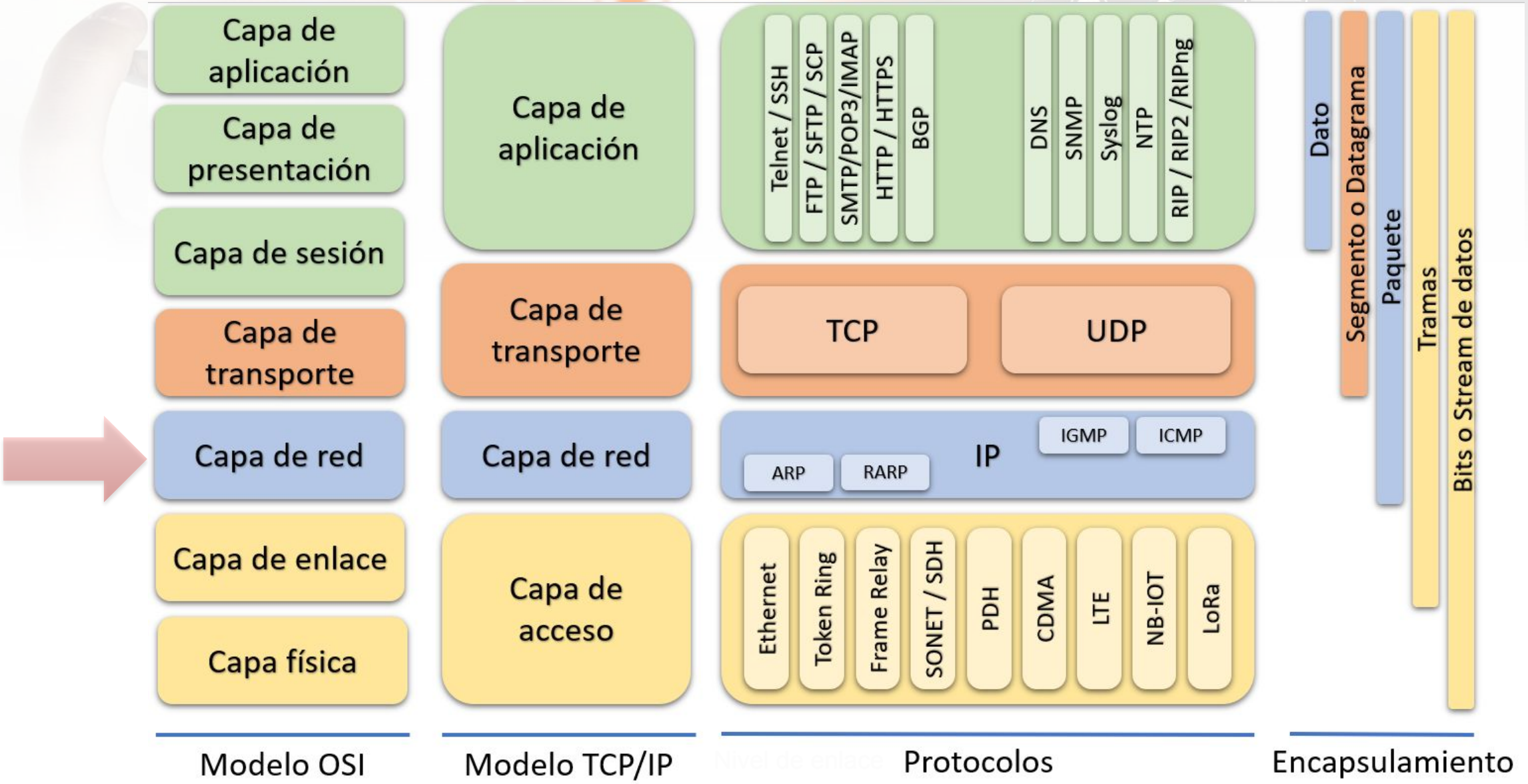
Capa de red

- Direccionamiento
- Circuitos virtuales
- Conmutación de paquetes
- IPv4
- Subredes
- DHCP
- DNAT / SNAT
- ICMP
- IPv6

Capa de red


Capa de aplicación
Capa de presentación
Capa de sesión
Capa de transporte
Capa de red
Capa de enlace
Capa física
Capa de aplicación
Capa de transporte
Capa de red
Capa de acceso
Telnet / SSH
FTP / SFTP / SCP
SMTP/POP3/IMAP
HTTP / HTTPS
BGP
DNS
SNMP
Syslog
NTP
RIP / RIP2 /RIPng
TCP
UDP
IP
IGMP
ICMP
ARP
RARP
Ethernet
Token Ring
Frame Relay
SONET / SDH
PDH
CDMA
LTE
NB-IOT
LoRa
Dato
Segmento o Datagrama
Paquete
Tramas
Bits o Stream de datos
Modelo OSI
Modelo TCP/IP
Protocolos
Encapsulamiento

Capa de red: servicios básicos

- Objetivo:

  Trasladar paquetes de un host a otro, que pueden estar en distintas redes.

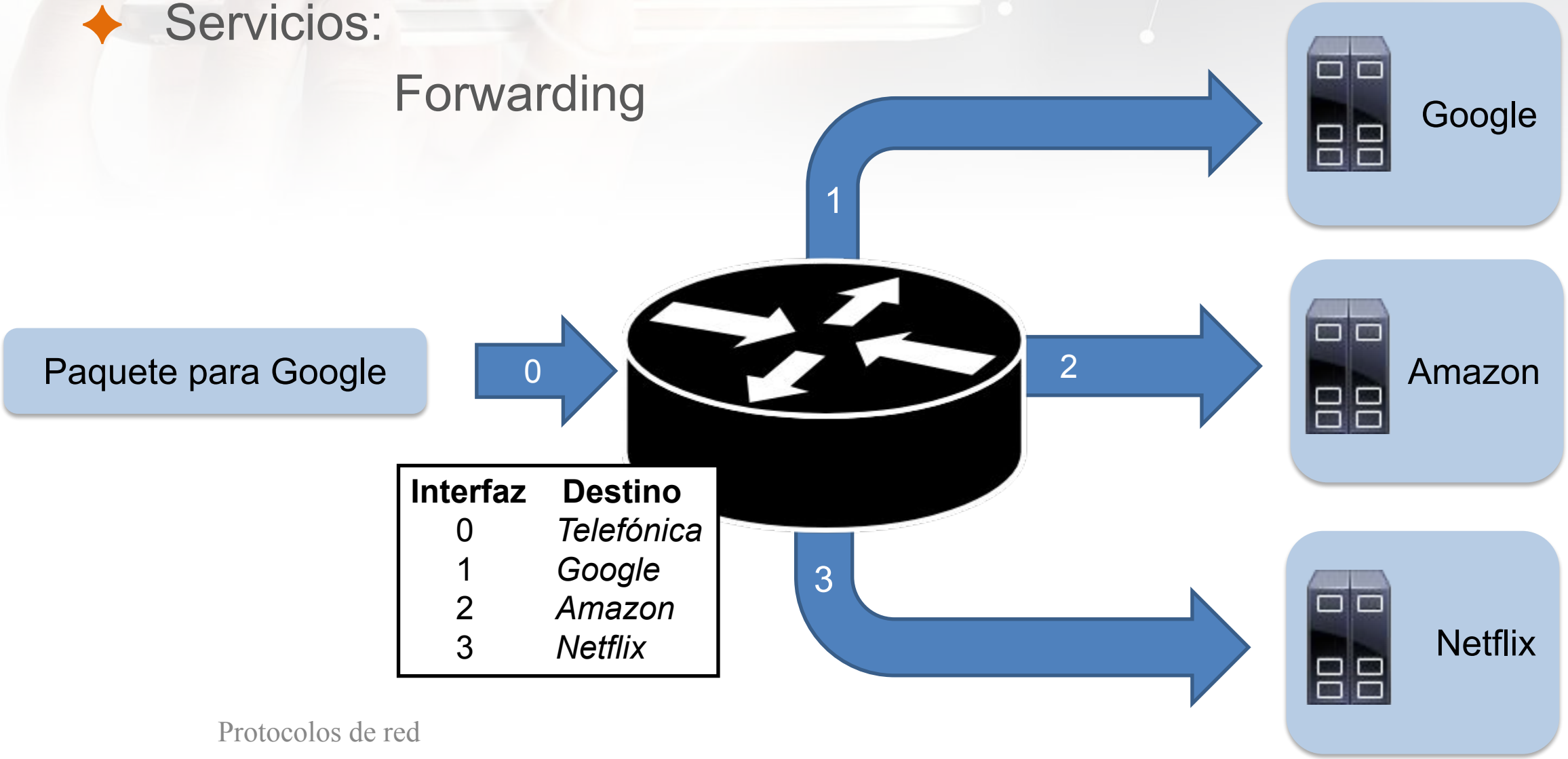
- Servicios:

  Encapsulamiento

Capa de red: servicios básicos

- Servicios:

  Forwarding

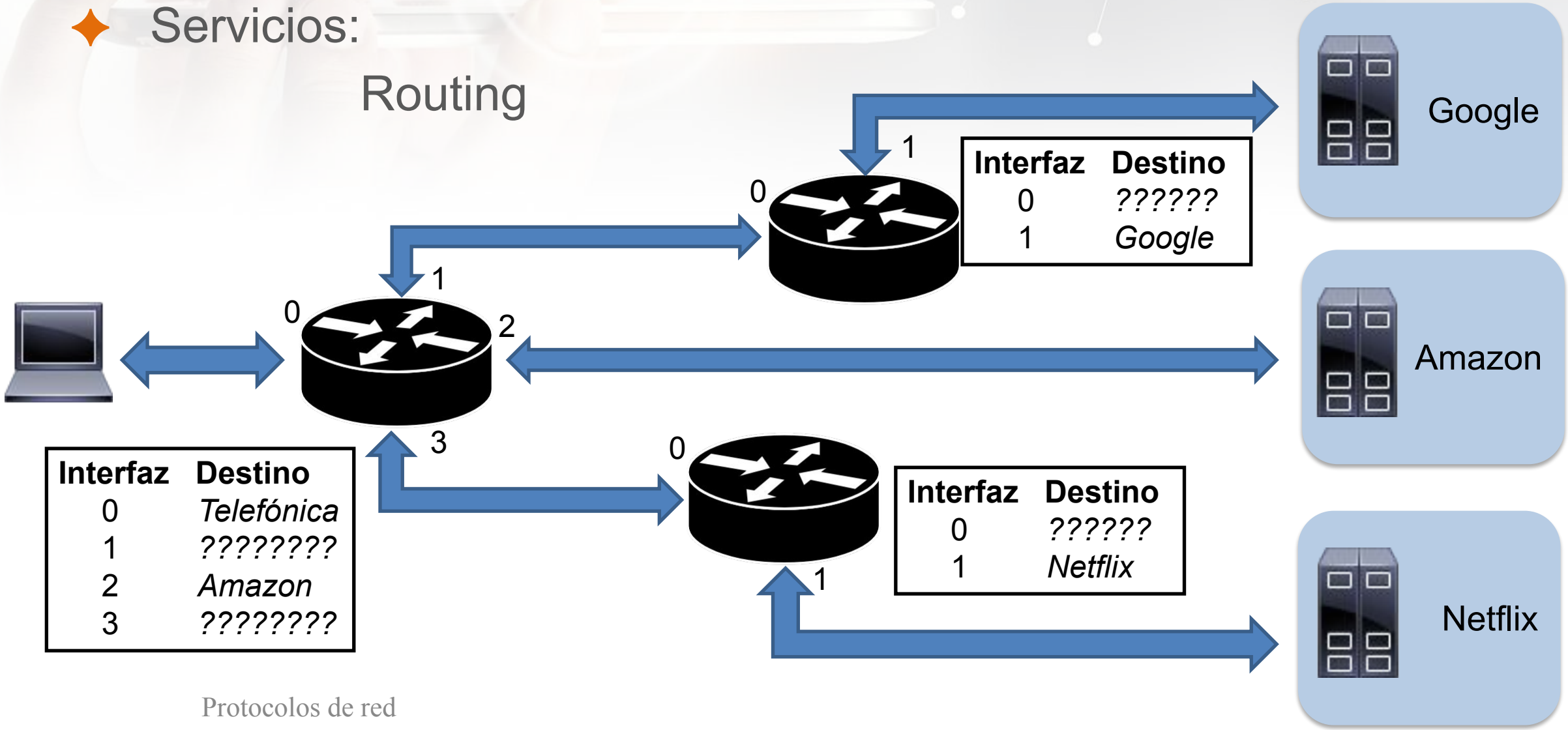


Interfaz	Destino
0	*Telefónica*
1	*Google*
2	*Amazon*
3	*Netflix*

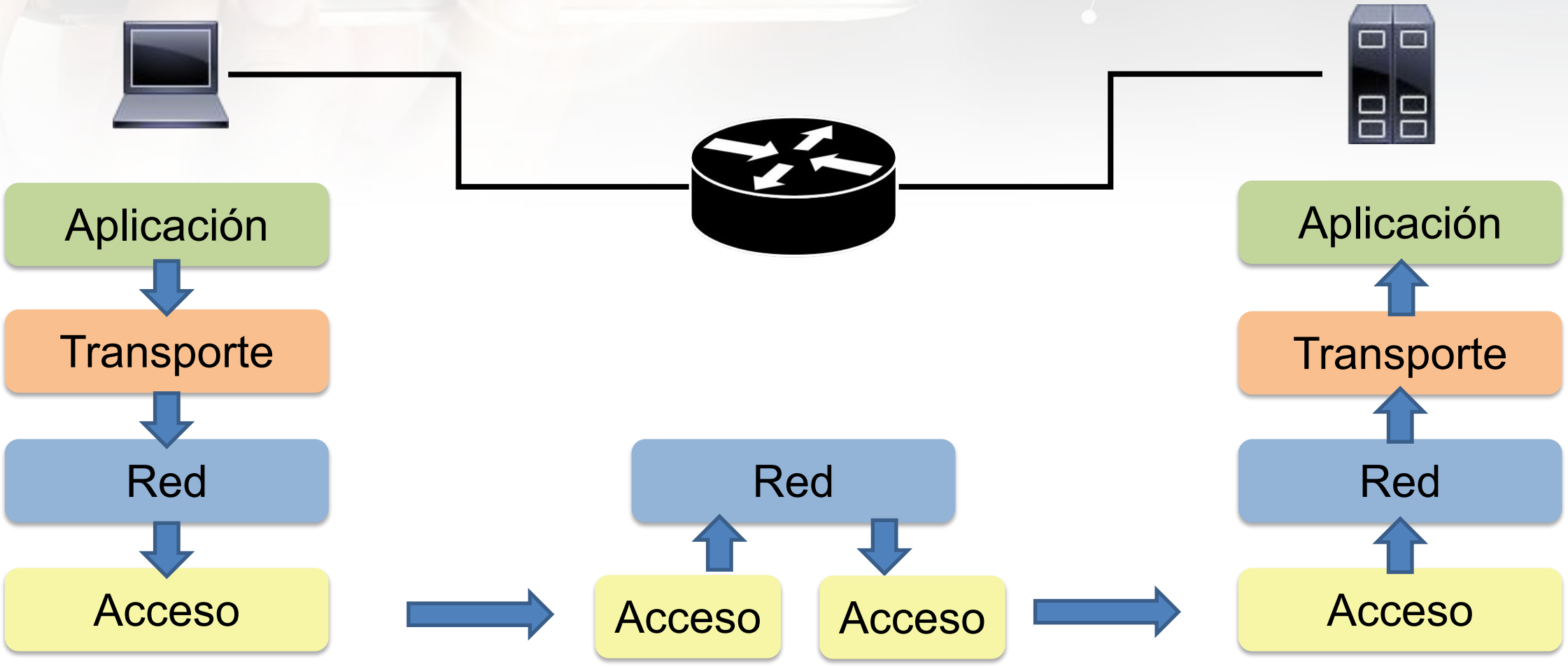
Capa de red: servicios básicos

- Servicios:

  Routing



Capa de red: servicios básicos

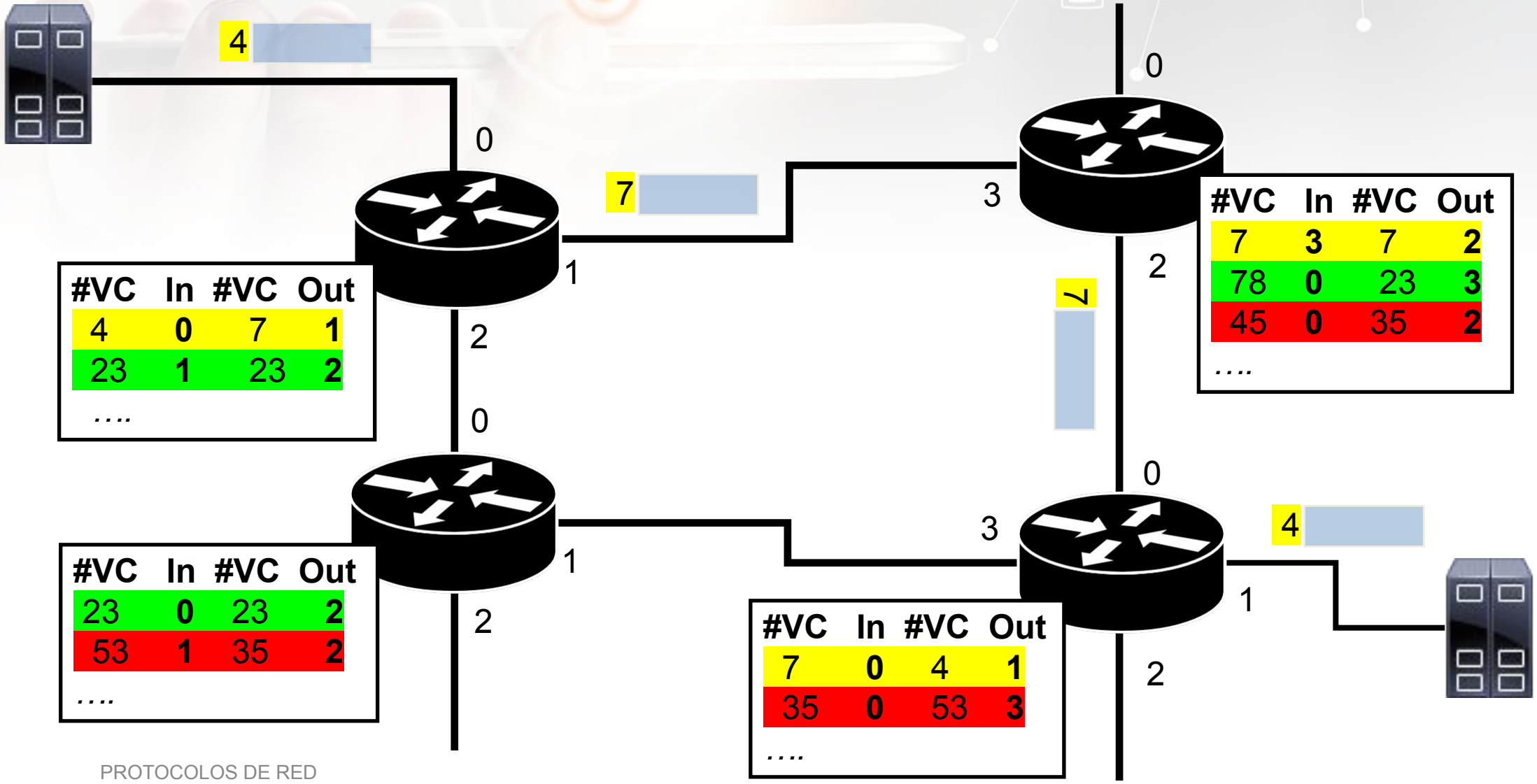


Capa de red: servicios adicionales

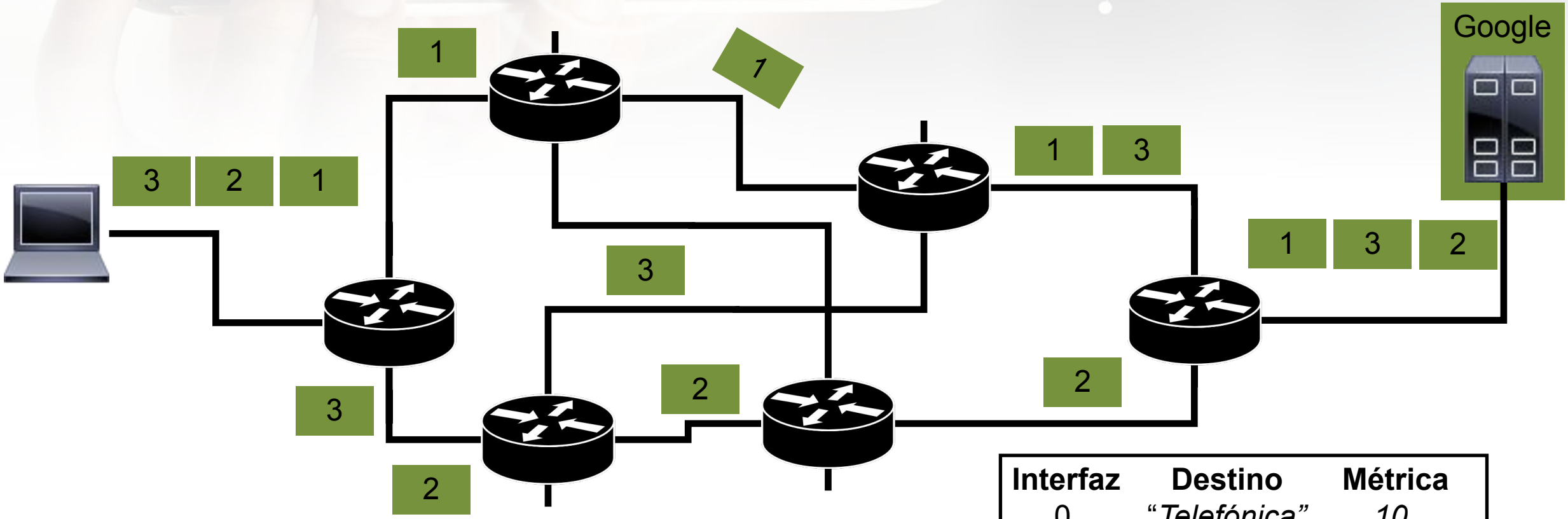
¿Qué servicios adicionales puede ofrecer un protocolo de capa de red?

- Entrega garantizada
- Demora mínima garantizada
- Orientado a Conexión
- Entrega en orden
- Ancho de banda mínimo
- Integridad de datos
- Encriptación

Virtual Circuits Networks



Datagram Networks



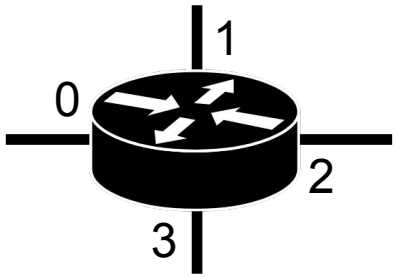
Interfaz	Destino	Métrica
0	*"Telefónica"*	*10*
1	*"Google"*	*10*
2	*"Amazon"*	*10*
3	*"Google*	*25*

Virtual Circuits vs Datagram Networks

	Datagrama	Circuito virtual
Establecer conexión	---	Requerido
Direccionamiento	Id. De host origen y destino	Número de CV
Info de estado	No	Tabla de subred con números de CV en cada router
Enrutamiento	Cada paquete una ruta independiente	Todos los paquetes del CV una misma ruta
¿Si falla un nodo?	Se pierden algunos paquetes	Todos los CV que pasan por el nodo finalizan
Control de congestión	Complejo	Simple
Complejidad	En la capa de transporte	En la capa de red

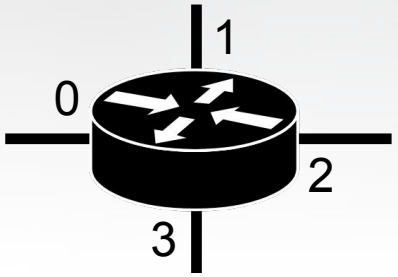
Capa de red: direcciones en tabla de ruteo

Rango de Direcciones	Interface
11001000 00010111 00010	0
11001000 00010111 00011000	1
11001000 00010111 00011	2
en otro caso	3



Capa de red: direcciones en tabla de ruteo

Rango de Direcciones	Interface
11001000 00010111 00010	0
11001000 00010111 00011000	1
11001000 00010111 00011	2
en otro caso	3



11001000 00010111 00010110 10100001

¿Por cual interface sale?

Interface 0

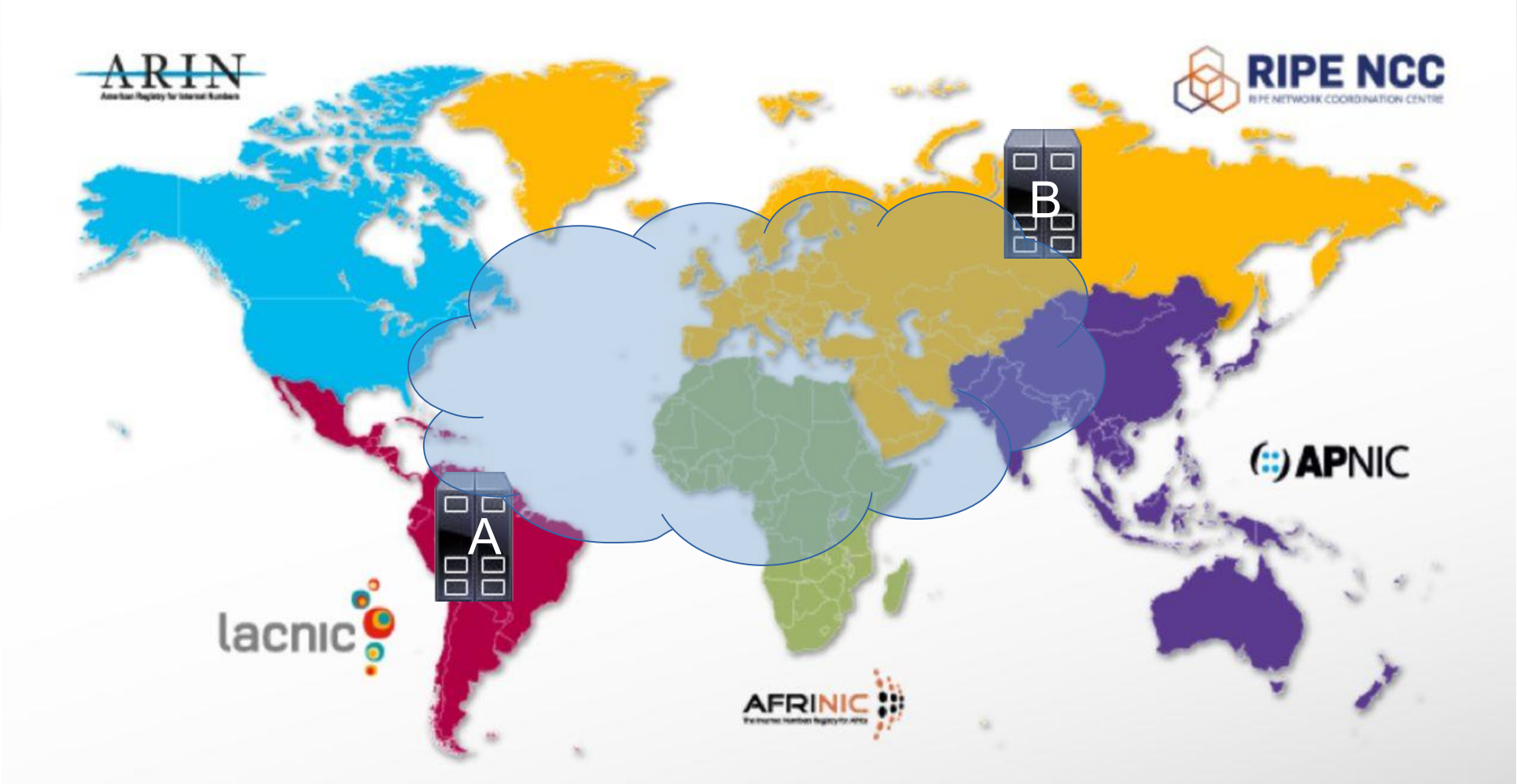
11001000 00010111 00011000 10101010

¿Sale por la interface 1 o 2 ?

Usando regla de coincidencia del prefijo más largo

Interface 1

Capa de red: direccionamiento

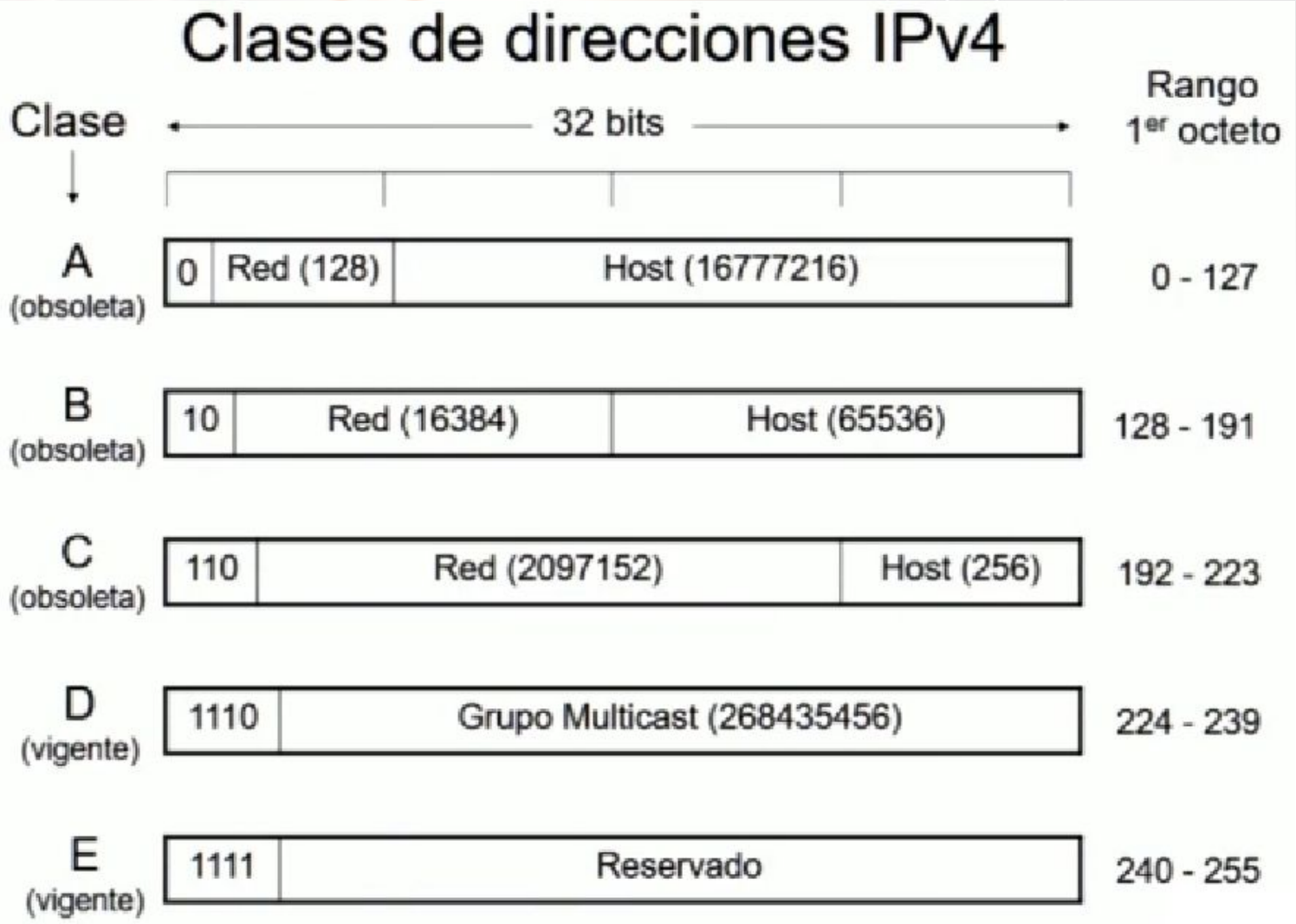


- 1972 *Jon Postel*
- 1976 *ISI* (Information Sciences Institute) Jon Postel
- *1988 IANA* (Internet Assigned Numbers Authority)
- 1993 *Regional Internet Registries (RIR)*

https://www.iana.org/numbers

Direccionamiento IPv4

RFC 790 (1981)



Direccionamiento IPv4

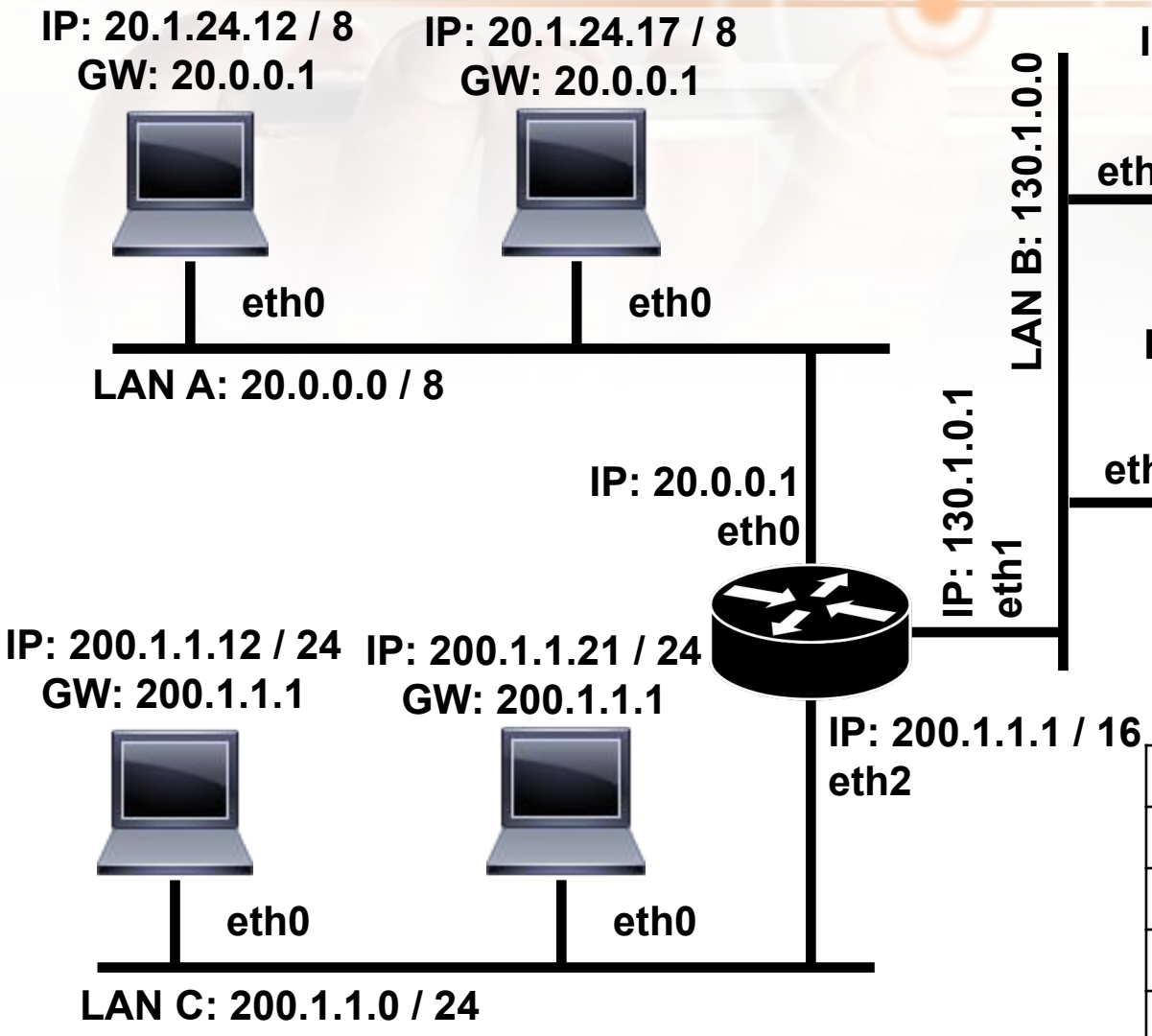
Direcciones IP reservadas

- **Loopback:** (RFC 990) en el año1989 :
  - las comenzadas con 127 (ej. 127.0.0.1)

- **Redes privadas** (RFC 1918) en el año 1996 :
  - 10.x.x.x
  - 172.16.0.0 a 172.31.255.255
  - 192.168.x.x

- **Link local** (RFC 3927) en el año 2005 :
  - 169.254.0.0 a 169.254.255.255

Direccionamiento IP



Internet (entre 1981 y 1993)

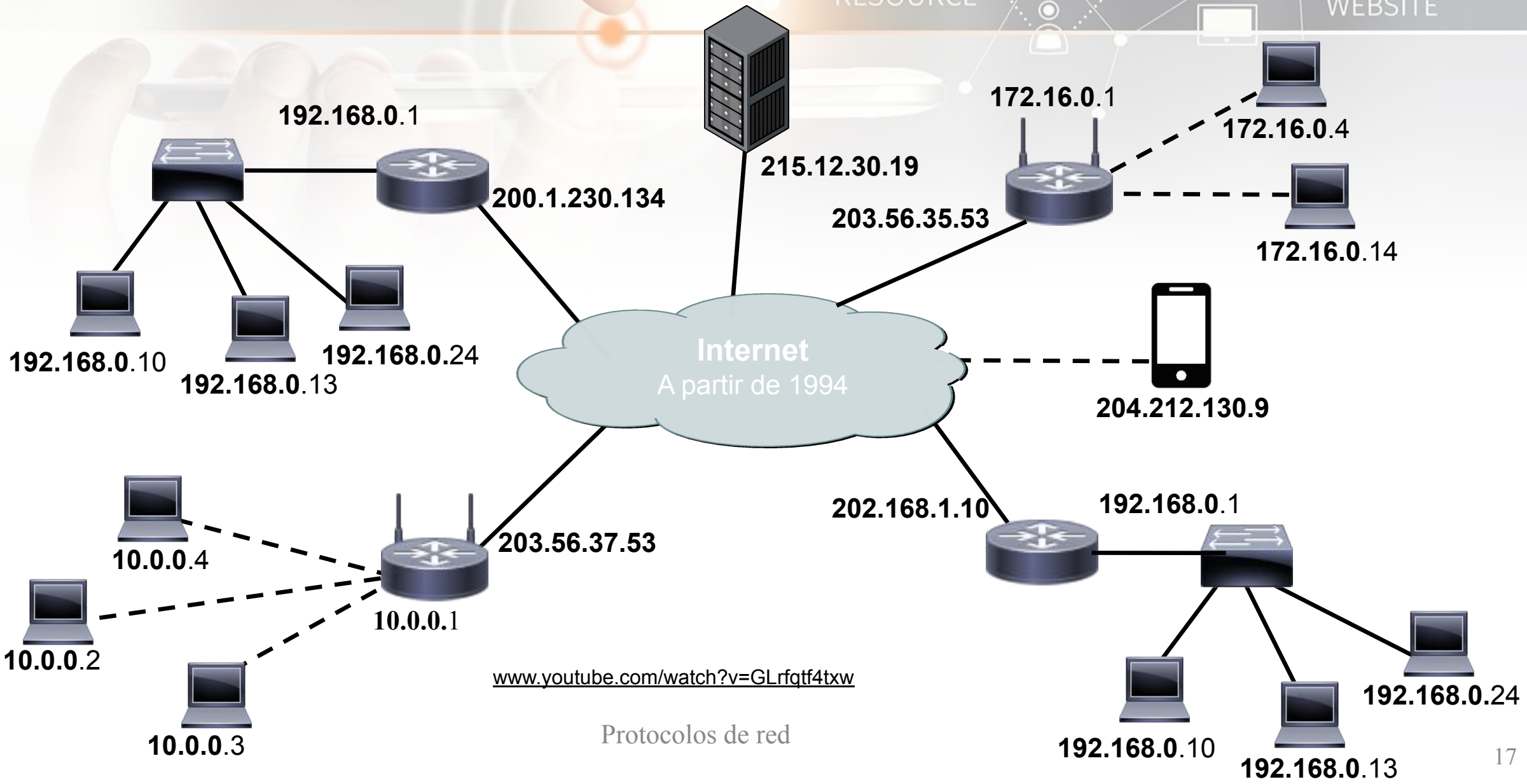
Router		
Destination	Gateway	Interface
20.0.0.0	0.0.0.0	eth0
130.1.0.0	0.0.0.0	eth1
200.1.1.0	0.0.0.0	eth2

Host @ LAN A		
Destination	Gateway	Interface
0.0.0.0	20.0.0.1	eth0
127.0.0.0	0.0.0.0	Lo
20.0.0.0	0.0.0.0	eth0

Hosts @ LAN B		
Destination	Gateway	Interface
0.0.0.0	130.1.0.1	eth0
127.0.0.0	0.0.0.0	Lo
130.1.0.0	0.0.0.0	eth0

Hosts @ LAN C		
Destination	Gateway	Interface
0.0.0.0	200.1.1.1	eth0
127.0.0.0	0.0.0.0	Lo
200.1.1.0	0.0.0.0	eth0

Direccionamiento IPv4



www.youtube.com/watch?v=GLrfqtf4txw

Subredes (VLSM) y Superred (CIDR)

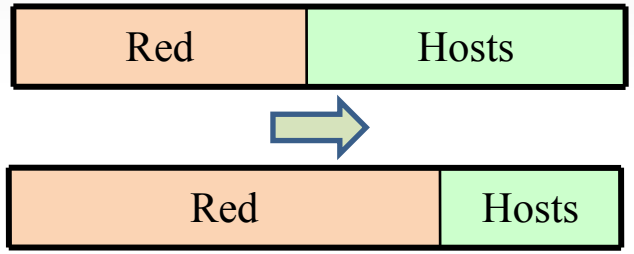
En 1993 se busca resolver dos grandes problemas

Agotamiento de direcciones IPv4

Crecimiento de las tablas de ruteo

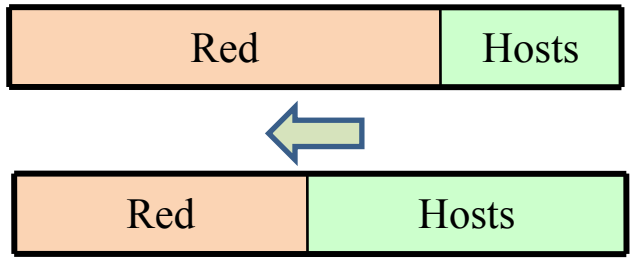
VLSM

Máscara de Subred de Longitud Variable



CIDR

Enrutamiento entre Dominios sin Clases



Subredes (VLSM)

Dada la red 192.168.80.0/24 se desea particionar en 4 subredes:

- **Guest 14 hosts** 2^**4** = 16
- **Robots 58 hosts** 2^**6** = 64
- **Servers 29 hosts** 2^**5** = 32
- **Workers 120 hosts** 2^**7** = 128

Mascara de Red Workers:

11111111 . 11111111 . 11111111 . **1**0000000

255 . 255 . 255 . 128

/ 25

Con 1 bit hay 2 redes posibles

Redes Disponibles		
Network Address	Usable Host Range	Broadcast Address:
Red Workers ➡ **192.168.80.0**	**192.168.80.1 - 192.168.80.126**	**192.168.80.127**
192.168.80.128	192.168.80.129 - 192.168.80.254	192.168.80.255

Subredes (VLSM)

Dada la red 192.168.80.0/24 se desea particionar en 4 subredes:

- **Guest 14 hosts** 2^**4** = 16
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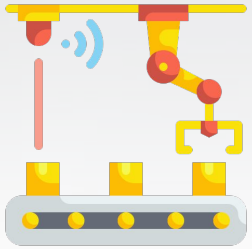


Mascara de Red Robots:

11111111 . 11111111 . 11111111 . **11**000000

255 . 255 . 255 . 192

/ 26

Con 2 bit hay 4 redes posibles

Redes Disponibles		
Network Address	**Usable Host Range**	**Broadcast Address:**
192.168.80.0	192.168.80.1 - 192.168.80.62	192.168.80.63
192.168.80.64	192.168.80.65 - 192.168.80.126	192.168.80.127
192.168.80.128	192.168.80.129 - 192.168.80.190	192.168.80.191
192.168.80.192	192.168.80.193 - 192.168.80.254	192.168.80.255

Red Workers

Red Robots

Subredes (VLSM)

Dada la red 192.168.80.0/24 se desea particionar en 4 subredes:

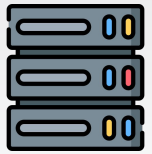
- **Guest 14 hosts** 2^**4** = 16
- **Robots 58 hosts** 2^**6** = 64
- **Servers 29 hosts** 2^**5** = 32
- **Workers 120 hosts** 2^**7** = 128

Mascara de Red Server:

11111111 . 11111111 . 11111111 . **111**00000

255 . 255 . 255 . 224

/ 27



Con 3 bit hay 8 redes posibles

	Redes Disponibles		
	Network Address	**Usable Host Range**	**Broadcast Address:**
Red Workers	192.168.80.0	192.168.80.1 - 192.168.80.30	192.168.80.31
	192.168.80.32	192.168.80.33 - 192.168.80.62	192.168.80.63
	192.168.80.64	192.168.80.65 - 192.168.80.94	192.168.80.95
	192.168.80.96	192.168.80.97 - 192.168.80.126	192.168.80.127
Red Robots	192.168.80.128	192.168.80.129 - 192.168.80.158	192.168.80.159
	192.168.80.160	192.168.80.161 - 192.168.80.190	192.168.80.191
Red Server	192.168.80.192	192.168.80.193 - 192.168.80.222	192.168.80.223
	192.168.80.224	192.168.80.225 - 192.168.80.254	192.168.80.255

Subredes (VLSM)

Dada la red 192.168.80.0/24 se desea particionar en 4 subredes:

- **Guest 14 hosts** 2^**4** = 16
- **Robots 58 hosts** 2^**6** = 64
- **Servers 29 hosts** 2^**5** = 32
- **Workers 120 hosts** 2^**7** = 128

Mascara de Red Guest:

11111111 . 11111111 . 11111111 . **1111**0000

255 . 255 . 255 . 240

/ 28

Con 4 bit hay 16 redes posibles

Redes Disponibles		
Network Address	Usable Host Range	Broadcast Address:
…	…	…
192.168.80.224	192.168.80.225 - 192.168.80.238	192.168.80.239
192.168.80.240	192.168.80.241 - 192.168.80.254	192.168.80.255

Red Guest ➡

Subredes (VLSM)

Motivos para crear subredes

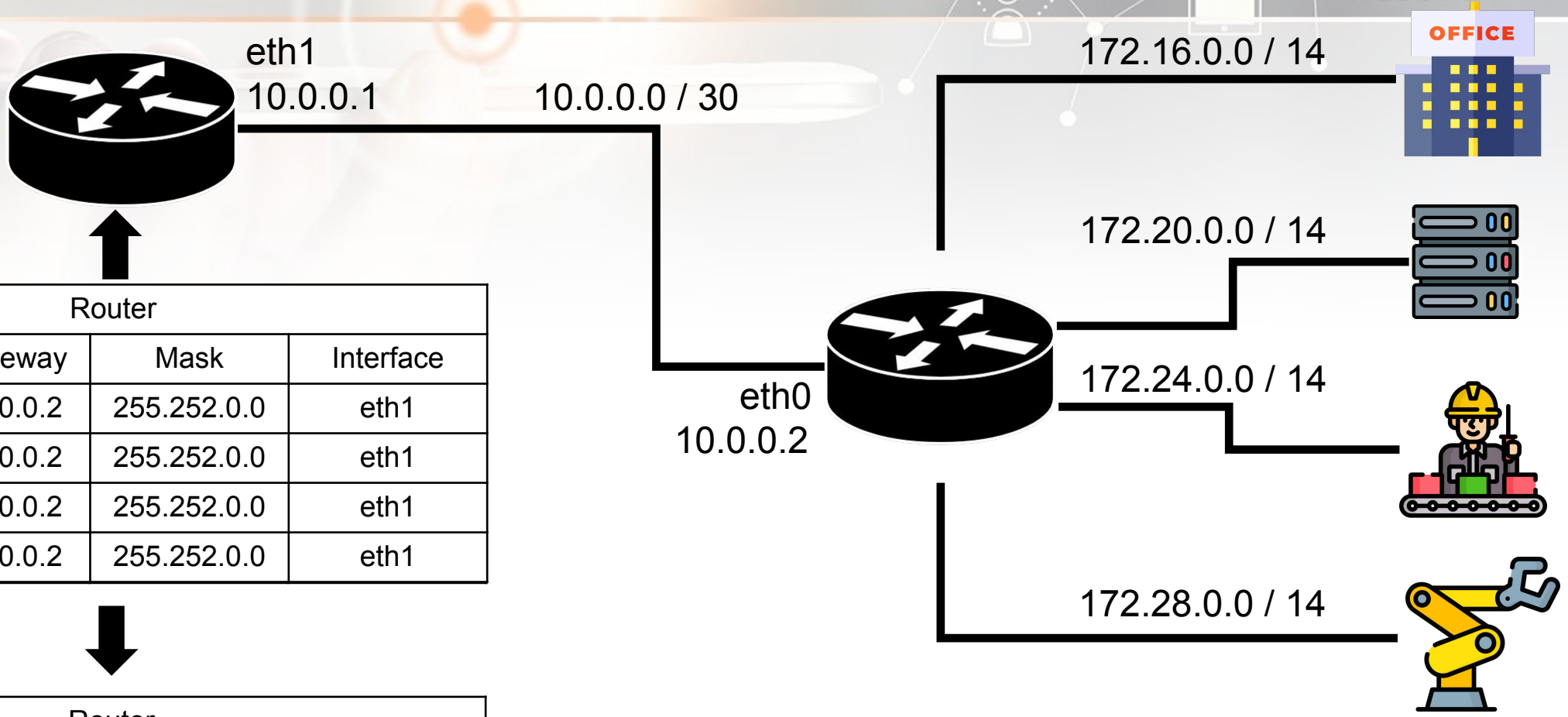
Razones topológicas

- Hosts muy distantes
- Interconectar distintas capas físicas
- Filtrar tráfico entre redes

Organización

- Simplificar la administración de la red
- Mapear la estructura de la organización
- Aislar el tráfico

Superred (CIDR)



Router			
Destination	Gateway	Mask	Interface
172.16.0.0	10.0.0.2	255.252.0.0	eth1
172.20.0.0	10.0.0.2	255.252.0.0	eth1
172.24.0.0	10.0.0.2	255.252.0.0	eth1
172.28.0.0	10.0.0.2	255.252.0.0	eth1

Router			
Destination	Gateway	Mask	Interface
172.16.0.0	10.0.0.2	255.240.0.0	eth1

252 = 1111 1100

16 = 0001 0000
20 = 0001 0100
24 = 0001 1000
28 = 0001 1100

252 = 1111 1100

240 = 1111 0000

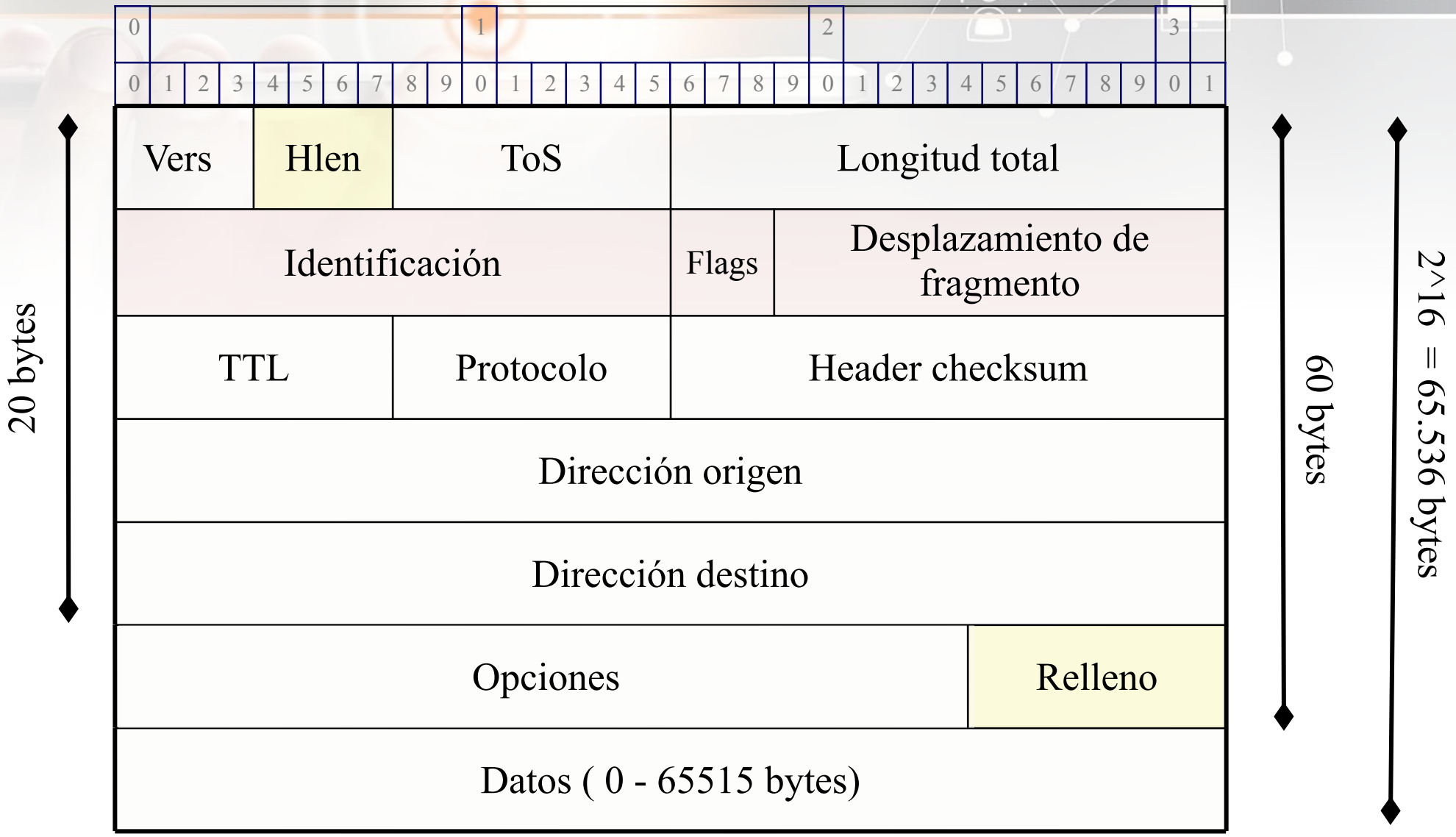
Superred (CIDR)

Motivos para crear superredes

- Optimización del Enrutamiento
- Escalabilidad
- Mejor uso del espacio de direcciones
- Reducción de latencia

Esto es implementable gracias a que la IANA a partir de 1993 empezó a segmentar en los RIR la asignación de bloques de direcciones.
Es decir las direcciones IP cobraron un orden jerárquico…

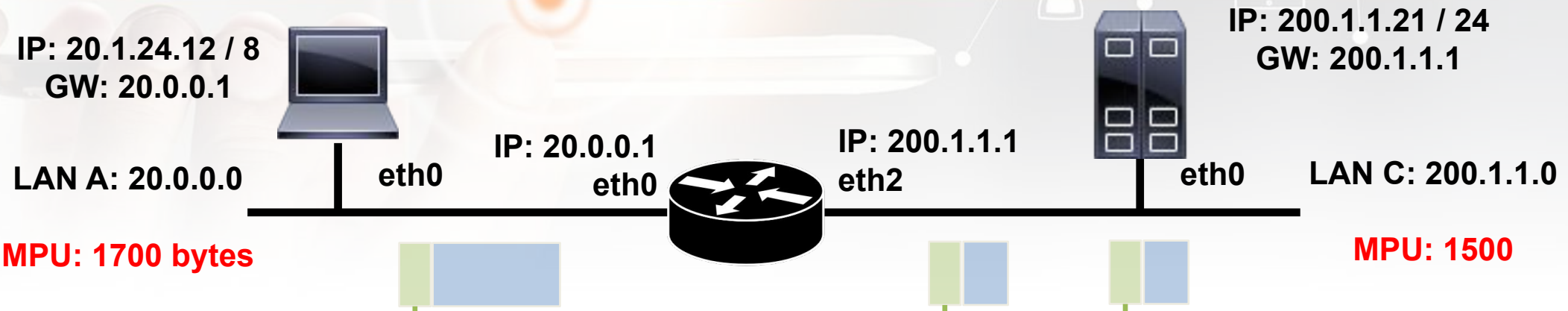
Paquete IPv4 (datagrama)



Paquete IPv4 (datagrama) - Protocolos

Valor	Protocolo	Descripción
1	ICMP	Internet Control Message Protocol
2	IGMP	Internet Group Management Protocol
3	GGP	Gateway-to-Gateway Protocol
4	IP	IP en IP (encapsulado)
5	ST	Stream
6	TCP	Transmission Control Protocol
8	EGP	Exterior Gateway Protocol
17	UDP	User Datagram Protocol
29	ISO-TP4	ISO Transport Protocol Clase 4
80	CLNP	Connectionless Network Protocol
88	IGRP/EIGRP	Interior Gateway Routing Protocol
89	OSPF	Open Shortest Path First

Fragmentación y ensamblado



Header IP			
Length	Id	Flags	Offset
1700	100	DF: 0 MF: 0	0

Header = 20 bytes
Payload = 1680 bytes

Header IP			
Length	Id	Flags	Offset
1500	100	DF: 0 MF: 1	0

Header = 20 bytes
Payload = 1480 bytes

Header IP			
Length	Id	Flags	Offset
220	100	DF: 0 MF: 0	185

Header = 20 bytes
Payload = 200 bytes

Asignación de direcciones IP

- Direccionamiento estático → Asignación Manual
- Direccionamiento dinámico → Local-Link
- Direccionamiento dinámico → RARP, BOOTP, DHCP

¿Qué información necesita una interfaz de red?

- Dirección IP
- Máscara de red
- Dirección del Gateway
- Direcciones de servidores DNS

Asignación de direcciones IP

1984
RARP
RFC 903

1985
BOOTP
RFC 951

1993
BOOTP
RFC 1532
RFC 1542

DHCP
RFC 1531
RFC 1541

1997
DHCP
RFC 2131
RFC 2132

2002
DHCP
RFC 3396
RFC 3442

2004
DHCP
RFC 3942

2013
DHCP
RFC 6842

Protocolo	Capa	Información Configurada	Limitaciones
RARP	Enlace	Solo dirección IP	Solo redes pequeñas.
BOOTP	Aplicación	Dirección IP y parámetros básicos	Sin asignación dinámica.
DHCP	Aplicación	Configuración completa	Depende de servidores DHCP.

DHCP (Dynamic Host Configuration Protocol)

RFC 2131 Dynamic Host Configuration Protocol March 1997

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|     op (1)    |   htype (1)   |   hlen (1)    |   hops (1)    |
+---------------+---------------+---------------+---------------+
|                            xid (4)                            |
+-------------------------------+-------------------------------+
|           secs (2)            |           flags (2)           |
+-------------------------------+-------------------------------+
|                          ciaddr  (4)                          |
+---------------------------------------------------------------+
|                          yiaddr  (4)                          |
+---------------------------------------------------------------+
|                          siaddr  (4)                          |
+---------------------------------------------------------------+
|                          giaddr  (4)                          |
+---------------------------------------------------------------+
|                                                               |
|                          chaddr  (16)                         |
|                                                               |
|                                                               |
+---------------------------------------------------------------+
|                                                               |
|                          sname   (64)                         |
+---------------------------------------------------------------+
|                                                               |
|                          file    (128)                        |
+---------------------------------------------------------------+
|                                                               |
|                          options (variable)                   |
+---------------------------------------------------------------+
```

Figure 1: Format of a DHCP message

FIELD	OCTETS	DESCRIPTION
op	1	Message op code / message type. 1 = BOOTREQUEST, 2 = BOOTREPLY
htype	1	Hardware address type, see ARP section in "Assigned Numbers" RFC; e.g., '1' = 10mb ethernet.
hlen	1	Hardware address length (e.g. '6' for 10mb ethernet).
hops	1	Client sets to zero, optionally used by relay agents when booting via a relay agent.
xid	4	Transaction ID, a random number chosen by the client, used by the client and server to associate messages and responses between a client and a server.
secs	2	Filled in by client, seconds elapsed since client began address acquisition or renewal process.
flags	2	Flags (see figure 2).
ciaddr	4	Client IP address; only filled in if client is in BOUND, RENEW or REBINDING state and can respond to ARP requests.
yiaddr	4	'your' (client) IP address.
siaddr	4	IP address of next server to use in bootstrap; returned in DHCPOFFER, DHCPACK by server.
giaddr	4	Relay agent IP address, used in booting via a relay agent.
chaddr	16	Client hardware address.
sname	64	Optional server host name, null terminated string.
file	128	Boot file name, null terminated string; "generic" name or null in DHCPDISCOVER, fully qualified directory-path name in DHCPOFFER.
options	var	Optional parameters field. See the options documents for a list of defined options.

Table 1: Description of fields in a DHCP message

DHCP (Dynamic Host Configuration Protocol)

Servidor DHCP
IP: 200.1.2.3
Puerto: 67

DHCP DISCOVER

src: 0.0.0.0 : 68
dest: 255.255.255.255 : 67
yiaddr: 0.0.0.0
transaction ID: 456

DHCP OFFER

src: 200.1.2.3 : 67
dest: 200.1.2.5 : 68
yiaddr: 200.1.2.5
transaction ID: 456
lifetime: 3600 secs

DHCP REQUEST

src: 0.0.0.0 : 68
dest: 255.255.255.255 : 67
yiaddr: 200.1.2.5
transaction ID: 456
lifetime: 3600 secs

DHCP ACK

src: 200.1.2.3 : 67
dest: 200.1.2.5 : 68
yiaddr: 200.1.2.5
transaction ID: 456
lifetime: 3600 secs

D. O. R. A.

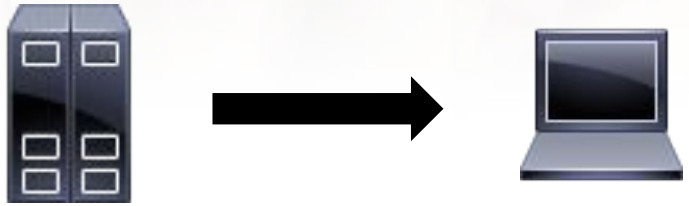
DHCP

DHCP DISCOVER

```
▷ Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
◢ User Datagram Protocol, Src Port: 68, Dst Port: 67
    Source Port: 68
    Destination Port: 67
    Length: 308
    Checksum: 0x61a0 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 0]
◢ Bootstrap Protocol (Discover)
    Message type: Boot Request (1)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0xfe089c15
    Seconds elapsed: 0
  ▷ Bootp flags: 0x0000 (Unicast)
    Client IP address: 0.0.0.0
    Your (client) IP address: 0.0.0.0
    Next server IP address: 0.0.0.0
    Relay agent IP address: 0.0.0.0
    Client MAC address: D-Link_12:47:cb (00:50:ba:12:47:cb)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
  ▷ Option: (53) DHCP Message Type (Discover)
  ▷ Option: (116) DHCP Auto-Configuration
  ▷ Option: (61) Client identifier
  ▷ Option: (50) Requested IP Address
  ▷ Option: (12) Host Name
  ▷ Option: (60) Vendor class identifier
  ▷ Option: (55) Parameter Request List
  ▷ Option: (255) End
    Padding: 00000000
```

DHCP

DHCP OFFER



Servidor DHCP
IP: 10.20.20.4
Puerto: 67

```
▷ Internet Protocol Version 4, Src: 10.20.20.4, Dst: 255.255.255.255
◢ User Datagram Protocol, Src Port: 67, Dst Port: 68
    Source Port: 67
    Destination Port: 68
    Length: 308
    Checksum: 0x8a44 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 2]
◢ Bootstrap Protocol (Offer)
    Message type: Boot Reply (2)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0xfe089c15
    Seconds elapsed: 0
  ▷ Bootp flags: 0x0000 (Unicast)
    Client IP address: 0.0.0.0
    Your (client) IP address: 10.20.20.20
    Next server IP address: 10.20.20.4
    Relay agent IP address: 0.0.0.0
    Client MAC address: D-Link_12:47:cb (00:50:ba:12:47:cb)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
  ▷ Option: (53) DHCP Message Type (Offer)
  ▷ Option: (54) DHCP Server Identifier
  ▷ Option: (51) IP Address Lease Time
  ▷ Option: (1) Subnet Mask
  ▷ Option: (15) Domain Name
  ▷ Option: (6) Domain Name Server
  ▷ Option: (255) End
    Padding: 000000000000000000000000000000000000
```

DHCP

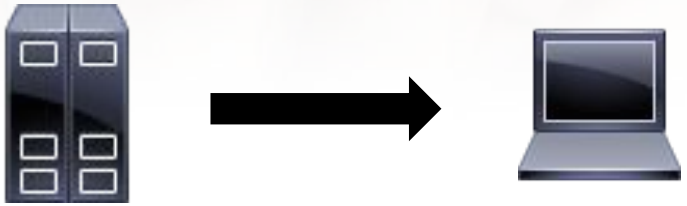
DHCP REQUEST

Servidor DHCP
IP: 10.20.20.4
Puerto: 67

```
▷ Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
◢ User Datagram Protocol, Src Port: 68, Dst Port: 67
    Source Port: 68
    Destination Port: 67
    Length: 322
    Checksum: 0xf62b [unverified]
    [Checksum Status: Unverified]
    [Stream index: 0]
◢ Bootstrap Protocol (Request)
    Message type: Boot Request (1)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0xfe089c15
    Seconds elapsed: 0
  ▷ Bootp flags: 0x0000 (Unicast)
    Client IP address: 0.0.0.0
    Your (client) IP address: 0.0.0.0
    Next server IP address: 0.0.0.0
    Relay agent IP address: 0.0.0.0
    Client MAC address: D-Link_12:47:cb (00:50:ba:12:47:cb)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
  ▷ Option: (53) DHCP Message Type (Request)
  ▷ Option: (61) Client identifier
  ▷ Option: (50) Requested IP Address
  ▷ Option: (54) DHCP Server Identifier
  ▷ Option: (12) Host Name
  ▷ Option: (81) Client Fully Qualified Domain Name
  ▷ Option: (60) Vendor class identifier
  ▷ Option: (55) Parameter Request List
  ▷ Option: (255) End
```

DHCP

DHCP ACK

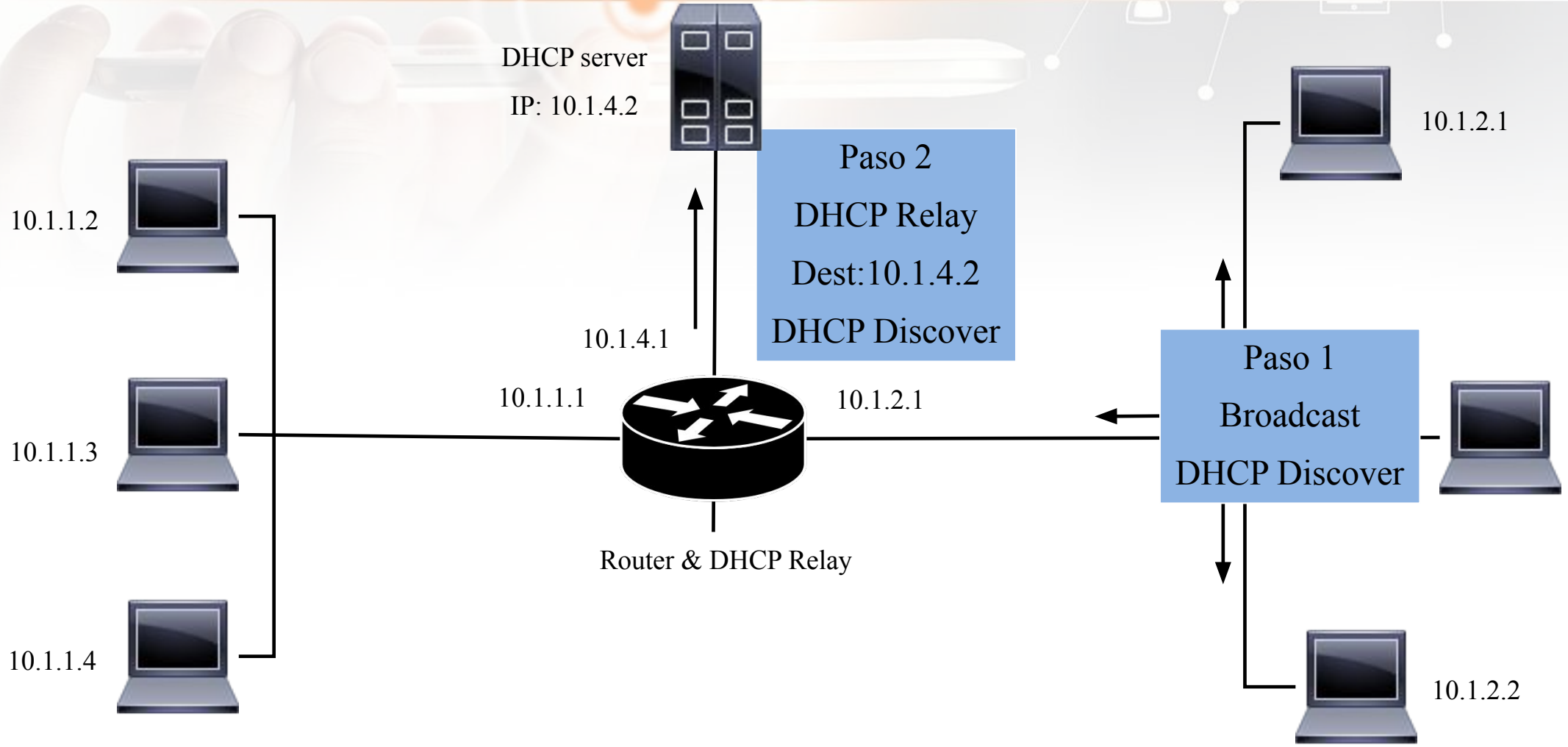


Servidor DHCP
IP: 10.20.20.4
Puerto: 67

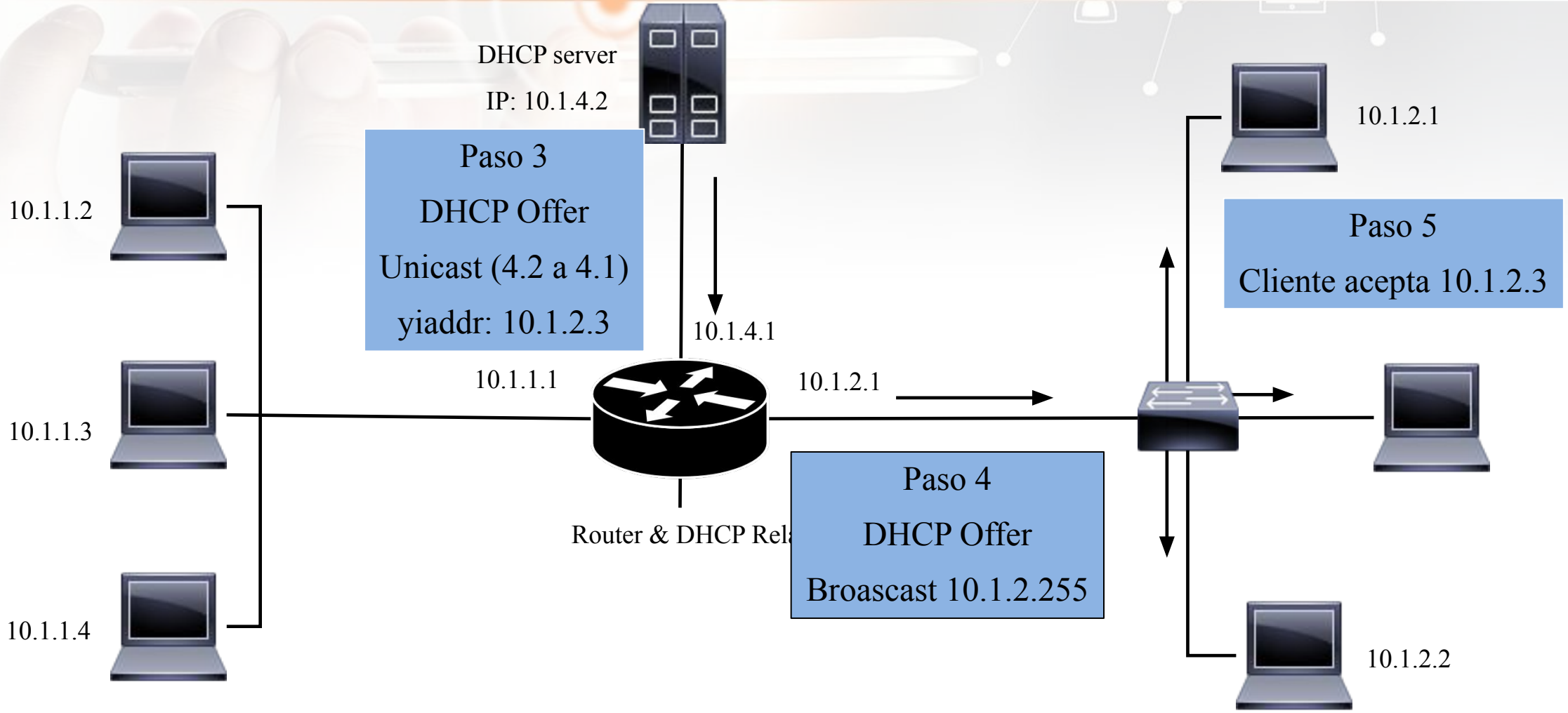
IP: 10.20.20.20

```
▷ Internet Protocol Version 4, Src: 10.20.20.4, Dst: 255.255.255.255
◢ User Datagram Protocol, Src Port: 67, Dst Port: 68
    Source Port: 67
    Destination Port: 68
    Length: 317
    Checksum: 0xb7a8 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 2]
◢ Bootstrap Protocol (ACK)
    Message type: Boot Reply (2)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0xfe089c15
    Seconds elapsed: 0
  ▷ Bootp flags: 0x0000 (Unicast)
    Client IP address: 0.0.0.0
    Your (client) IP address: 10.20.20.20
    Next server IP address: 10.20.20.4
    Relay agent IP address: 0.0.0.0
    Client MAC address: D-Link_12:47:cb (00:50:ba:12:47:cb)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
  ▷ Option: (53) DHCP Message Type (ACK)
  ▷ Option: (54) DHCP Server Identifier
  ▷ Option: (51) IP Address Lease Time
  ▷ Option: (81) Client Fully Qualified Domain Name
  ▷ Option: (1) Subnet Mask
  ▷ Option: (15) Domain Name
  ▷ Option: (6) Domain Name Server
  ▷ Option: (255) End
```

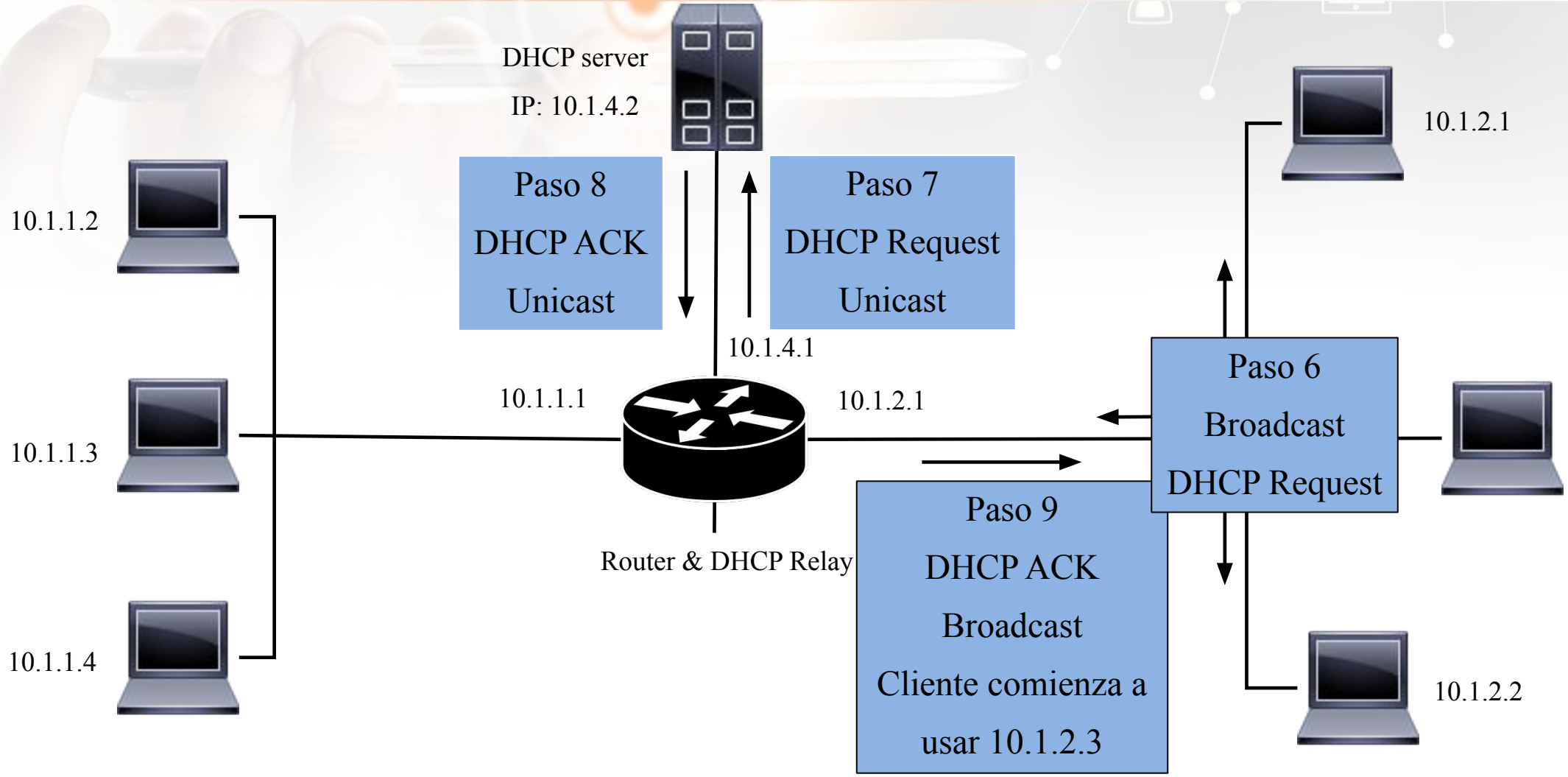
DHCP: Relay Agent



DHCP: Relay Agent



DHCP: Relay Agent



DHCP: renew request

```
▷ Internet Protocol Version 4, Src: 192.168.1.101, Dst: 192.168.1.1
▷ User Datagram Protocol, Src Port: 68, Dst Port: 67
◢ Bootstrap Protocol (Request)
    Message type: Boot Request (1)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0x0dd1b124
    Seconds elapsed: 0
  ▷ Bootp flags: 0x0000 (Unicast)
    Client IP address: 192.168.1.101
    Your (client) IP address: 0.0.0.0
    Next server IP address: 0.0.0.0
    Relay agent IP address: 0.0.0.0
    Client MAC address: AsustekC_22:77:a0 (40:16:7e:22:77:a0)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
  ▷ Option: (53) DHCP Message Type (Request)
  ◢ Option: (61) Client identifier
      Length: 7
      Hardware type: Ethernet (0x01)
      Client MAC address: AsustekC_22:77:a0 (40:16:7e:22:77:a0)
```

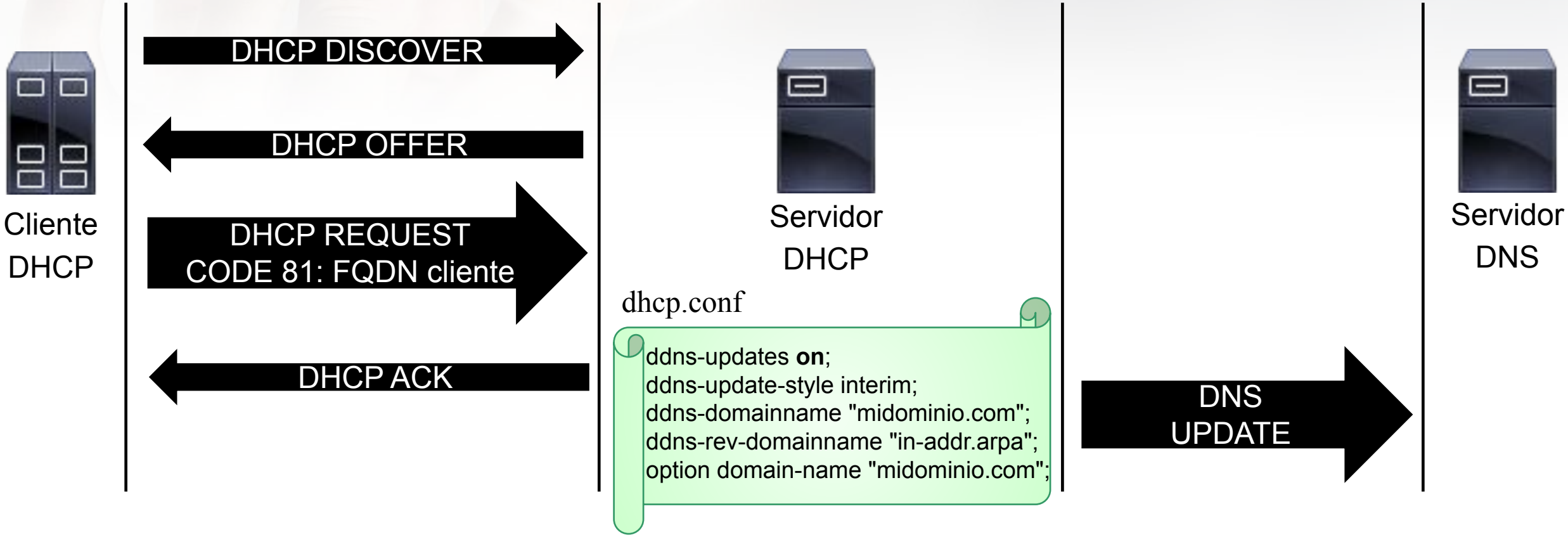
```
  ▷ Option: (12) Host Name
  ▷ Option: (81) Client Fully Qualified Domain Name
  ▷ Option: (60) Vendor class identifier
  ◢ Option: (55) Parameter Request List
      Length: 12
      Parameter Request List Item: (1) Subnet Mask
      Parameter Request List Item: (15) Domain Name
      Parameter Request List Item: (3) Router
      Parameter Request List Item: (6) Domain Name Server
      Parameter Request List Item: (44) NetBIOS over TCP/IP Name Server
      Parameter Request List Item: (46) NetBIOS over TCP/IP Node Type
      Parameter Request List Item: (47) NetBIOS over TCP/IP Scope
      Parameter Request List Item: (31) Perform Router Discover
      Parameter Request List Item: (33) Static Route
      Parameter Request List Item: (121) Classless Static Route
      Parameter Request List Item: (249) Private/Classless Static Route (Microsoft)
      Parameter Request List Item: (43) Vendor-Specific Information
  ▷ Option: (255) End
```

DHCP: renew ACK

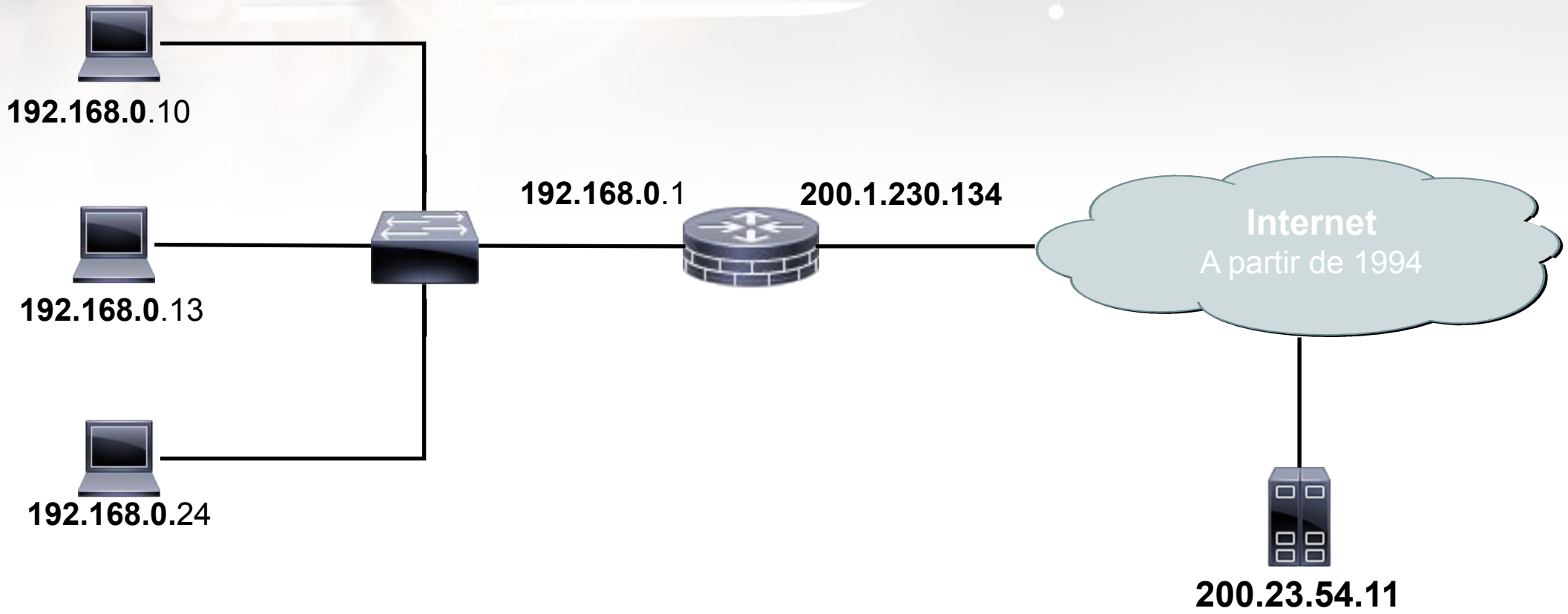
```
▷ Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.101
▷ User Datagram Protocol, Src Port: 67, Dst Port: 68
◢ Bootstrap Protocol (ACK)
    Message type: Boot Reply (2)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0x0dd1b124
    Seconds elapsed: 0
  ▷ Bootp flags: 0x0000 (Unicast)
    Client IP address: 192.168.1.101
    Your (client) IP address: 192.168.1.101
    Next server IP address: 0.0.0.0
    Relay agent IP address: 0.0.0.0
    Client MAC address: AsustekC_22:77:a0 (40:16:7e:22:77:a0)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
```

```
◢ Option: (53) DHCP Message Type (ACK)
      Length: 1
      DHCP: ACK (5)
◢ Option: (54) DHCP Server Identifier
      Length: 4
      DHCP Server Identifier: 192.168.1.1
◢ Option: (51) IP Address Lease Time
      Length: 4
      IP Address Lease Time: (4294967295s) infinity
◢ Option: (1) Subnet Mask
      Length: 4
      Subnet Mask: 255.255.255.0
◢ Option: (3) Router
      Length: 4
      Router: 192.168.1.1
◢ Option: (6) Domain Name Server
      Length: 8
      Domain Name Server: 208.67.222.222
      Domain Name Server: 8.8.8.8
◢ Option: (255) End
      Option End: 255
  Padding: 000000000000000000000000000000000000000000000000
```

DNS dinámico, integrado con DHCP

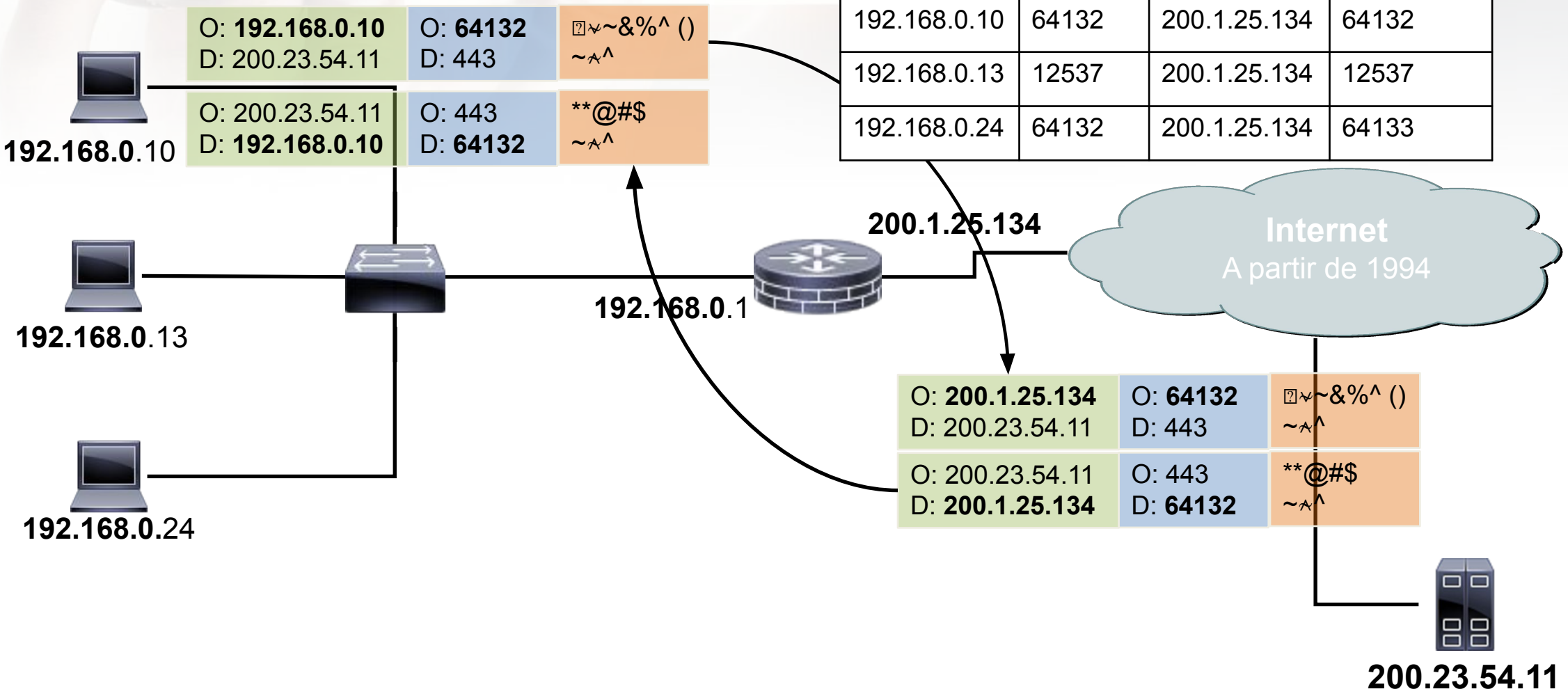


Network Address Translation (NAT)



Source NAT (SNAT)

IP Privada	Puerto Privado	IP Pública	Puerto Público
192.168.0.10	64132	200.1.25.134	64132
192.168.0.13	12537	200.1.25.134	12537
192.168.0.24	64132	200.1.25.134	64133

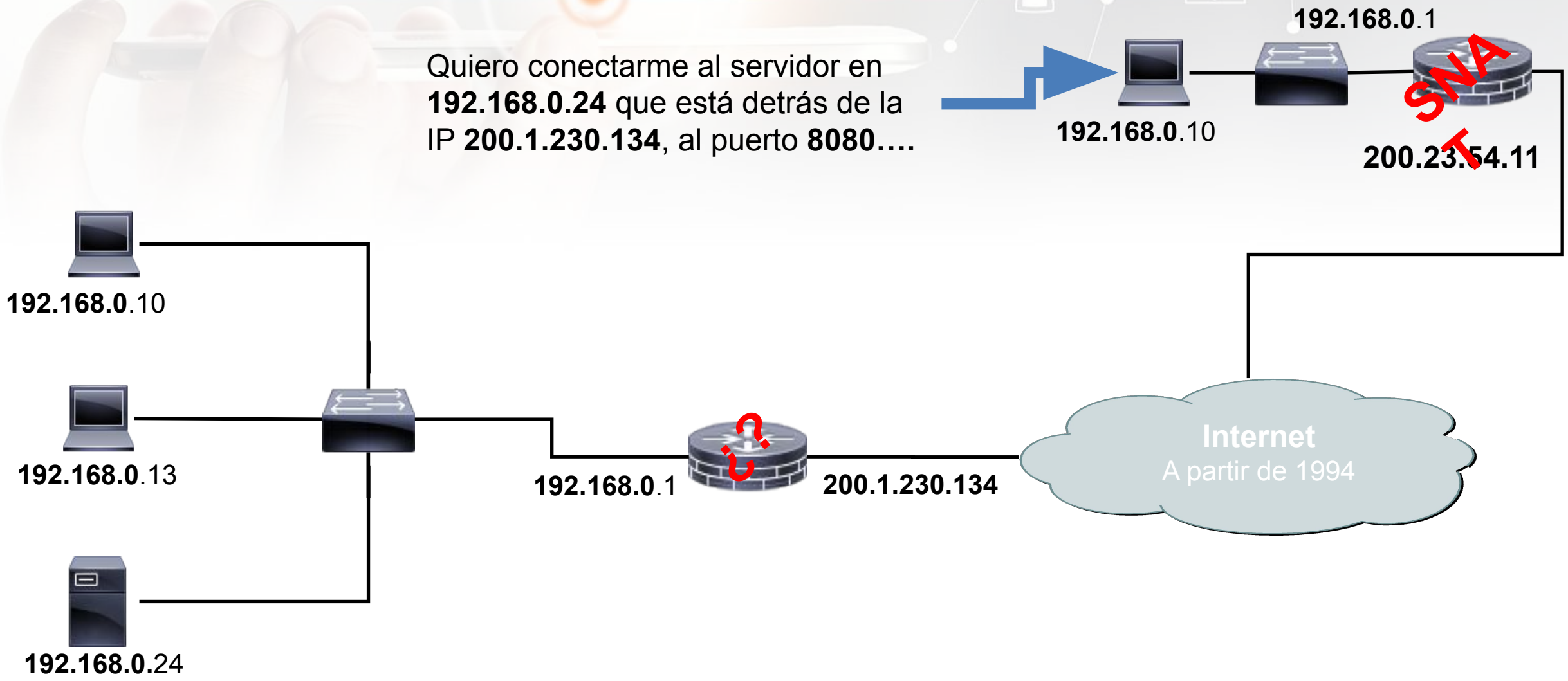


Source NAT (SNAT)

¿Qué campos se modifican al aplicar SNAT a un paquete IP?

<table>
<tr><td>Vers</td><td>Hlen</td><td>ToS</td><td colspan="2">Longitud total</td></tr>
<tr><td colspan="3">Identificación</td><td>Flags</td><td>Desplazam. de fragmento</td></tr>
<tr><td colspan="2">TTL</td><td>Protocolo</td><td colspan="2">Header checksum</td></tr>
<tr><td colspan="5">Dirección origen</td></tr>
<tr><td colspan="5">Dirección destino</td></tr>
<tr><td colspan="4">Opciones</td><td>Relleno</td></tr>
<tr><td colspan="5">Datos</td></tr>
</table>

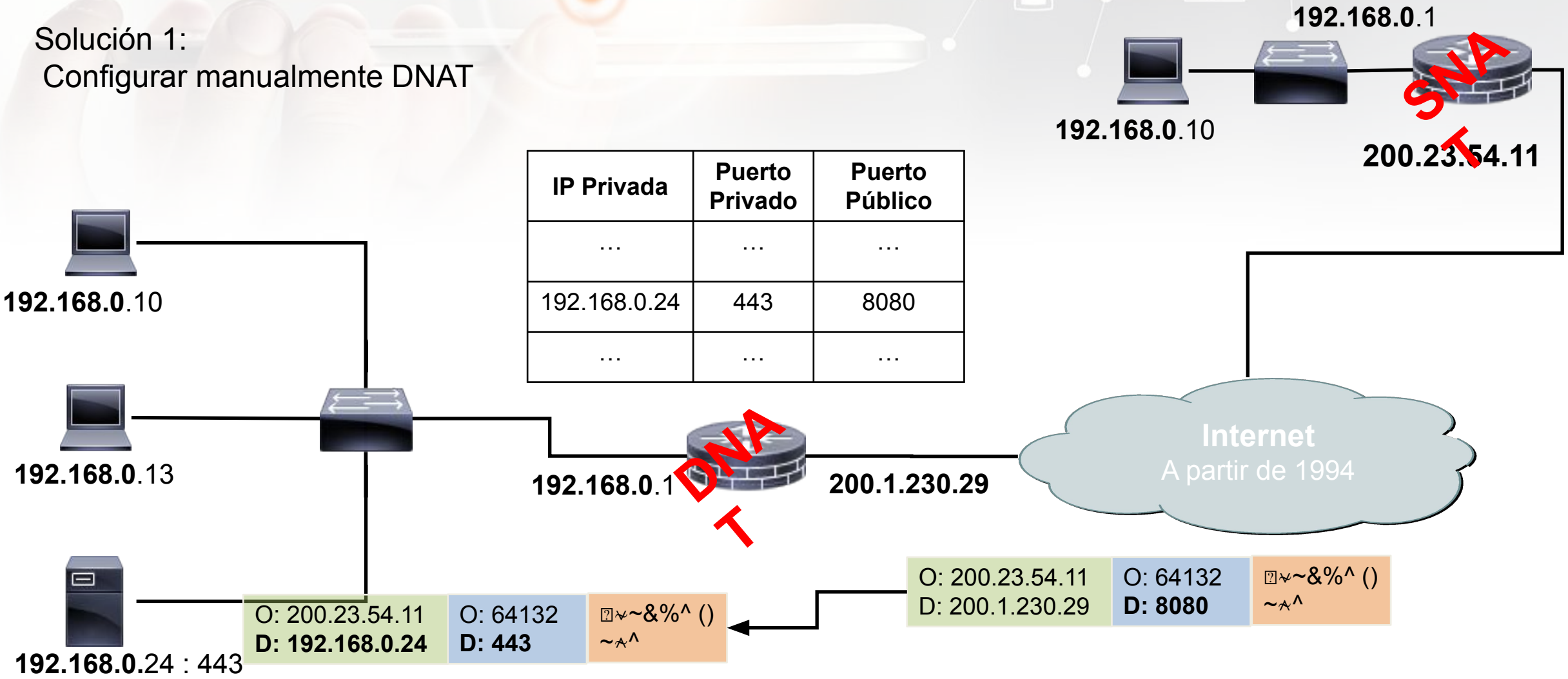
Destination NAT (port forwading)



Destination NAT (port forwading)

Solución 1:
Configurar manualmente DNAT

IP Privada	Puerto Privado	Puerto Público
…	…	…
192.168.0.24	443	8080
…	…	…



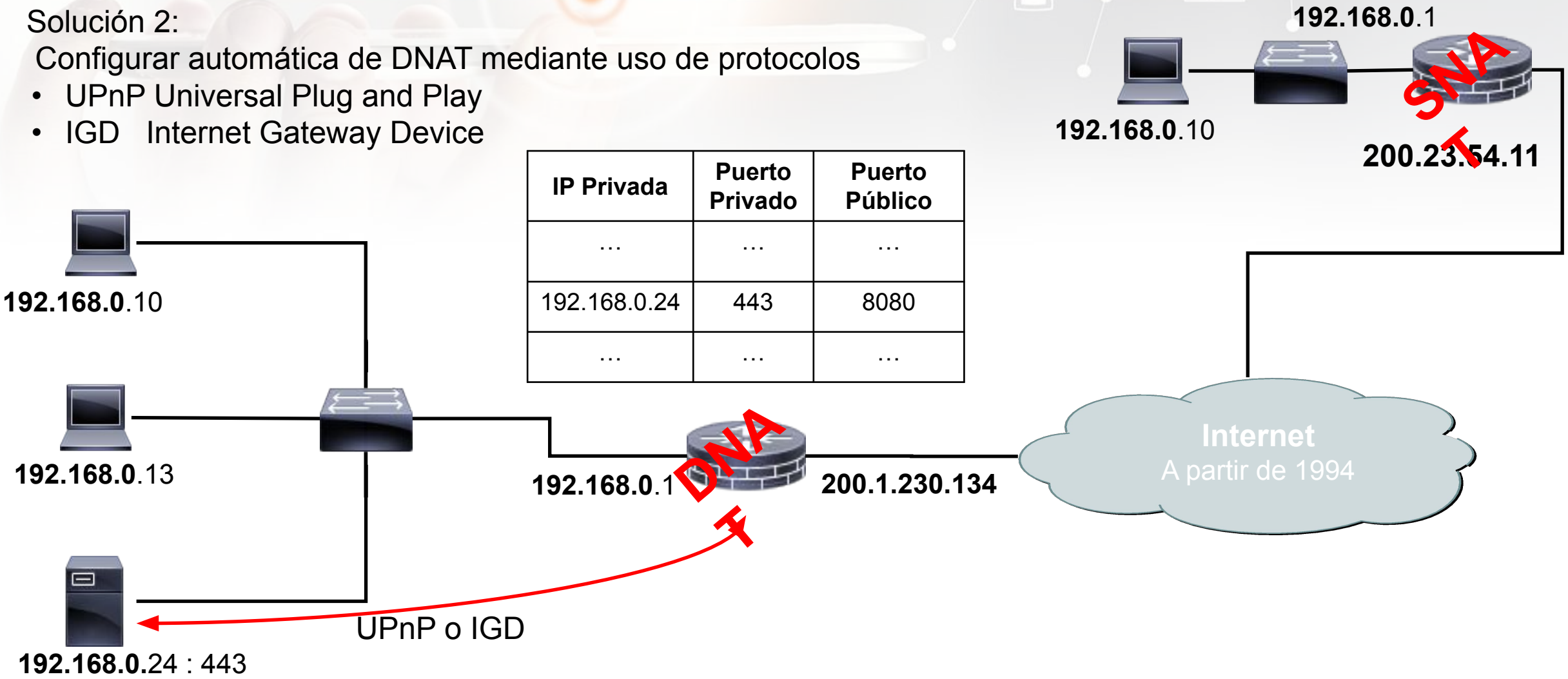
Destination NAT (port forwading)

Solución 2:

Configurar automática de DNAT mediante uso de protocolos

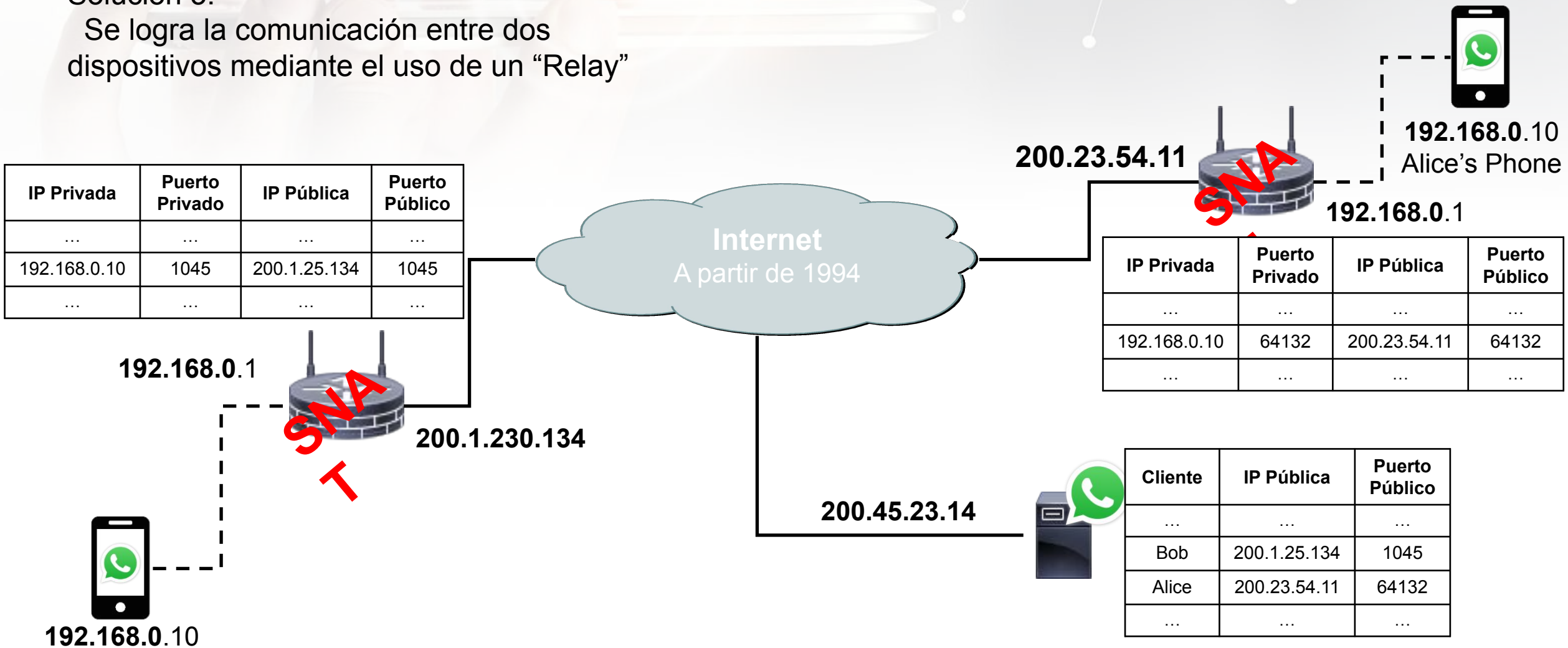
- UPnP Universal Plug and Play
- IGD Internet Gateway Device

IP Privada	Puerto Privado	Puerto Público
…	…	…
192.168.0.24	443	8080
…	…	…



NAT

Solución 3:
Se logra la comunicación entre dos dispositivos mediante el uso de un “Relay”



IP Privada	Puerto Privado	IP Pública	Puerto Público
…	…	…	…
192.168.0.10	1045	200.1.25.134	1045
…	…	…	…

IP Privada	Puerto Privado	IP Pública	Puerto Público
…	…	…	…
192.168.0.10	64132	200.23.54.11	64132
…	…	…	…

Cliente	IP Pública	Puerto Público
…	…	…
Bob	200.1.25.134	1045
Alice	200.23.54.11	64132
…	…	…

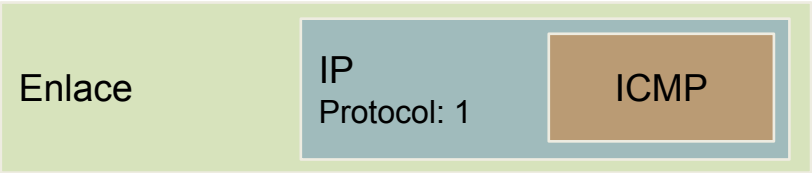
NAT

=

ICMP

0								1								2															
0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1
Type								Code								Checksum															
Data (Depende de Código y Dato)																															

Los valores para los campos Tipo y Código se definen en RFC1700



ICMP : Timestamp request / reply

```
                                                          September 1981
RFC 792


Timestamp or Timestamp Reply Message

    0                   1                   2                   3
    0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
   +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
   |     Type      |     Code      |          Checksum             |
   +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
   |           Identifier          |        Sequence Number        |
   +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
   |     Originate Timestamp                                       |
   +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
   |     Receive Timestamp                                         |
   +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
   |     Transmit Timestamp                                        |
   +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

```
Type

   13 for timestamp message;

   14 for timestamp reply message.

Code

   0
```

ICMP : Destination Unreachable Message

```
                                                          September 1981
RFC 792


Destination Unreachable Message

    0                   1                   2                   3
    0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
   +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
   |     Type      |     Code      |          Checksum             |
   +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
   |                             unused                            |
   +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
   |      Internet Header + 64 bits of Original Data Datagram      |
   +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+

   IP Fields:

   Destination Address

      The source network and address from the original datagram's data.

   ICMP Fields:

   Type

      3

   Code

      0 = net unreachable;

      1 = host unreachable;

      2 = protocol unreachable;

      3 = port unreachable;

      4 = fragmentation needed and DF set;

      5 = source route failed.
```

```
RFC 1700                    Assigned Numbers                  October 1994


   2     Unassigned                                              [JBP]

   3     Destination Unreachable                              [RFC792]

         Codes
             0  Net Unreachable
             1  Host Unreachable
             2  Protocol Unreachable
             3  Port Unreachable
             4  Fragmentation Needed and Don't Fragment was Set
             5  Source Route Failed
             6  Destination Network Unknown
             7  Destination Host Unknown
             8  Source Host Isolated
             9  Communication with Destination Network is
                Administratively Prohibited
            10  Communication with Destination Host is
                Administratively Prohibited
            11  Destination Network Unreachable for Type of Service
            12  Destination Host Unreachable for Type of Service

Internet Header + 64 bits of Data Datagram

   The internet header plus the first 64 bits of the original
   datagram's data.  This data is used by the host to match the
   message to the appropriate process.  If a higher level protocol
   uses port numbers, they are assumed to be in the first 64 data
   bits of the original datagram's data.
```

ICMP : Redirect Message

September 1981

RFC 792

Redirect Message

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|     Type      |     Code      |          Checksum             |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                 Gateway Internet Address                      |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|      Internet Header + 64 bits of Original Data Datagram      |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

Type

5

Code

0 = Redirect datagrams for the Network.

1 = Redirect datagrams for the Host.

2 = Redirect datagrams for the Type of Service and Network.

3 = Redirect datagrams for the Type of Service and Host.

Checksum

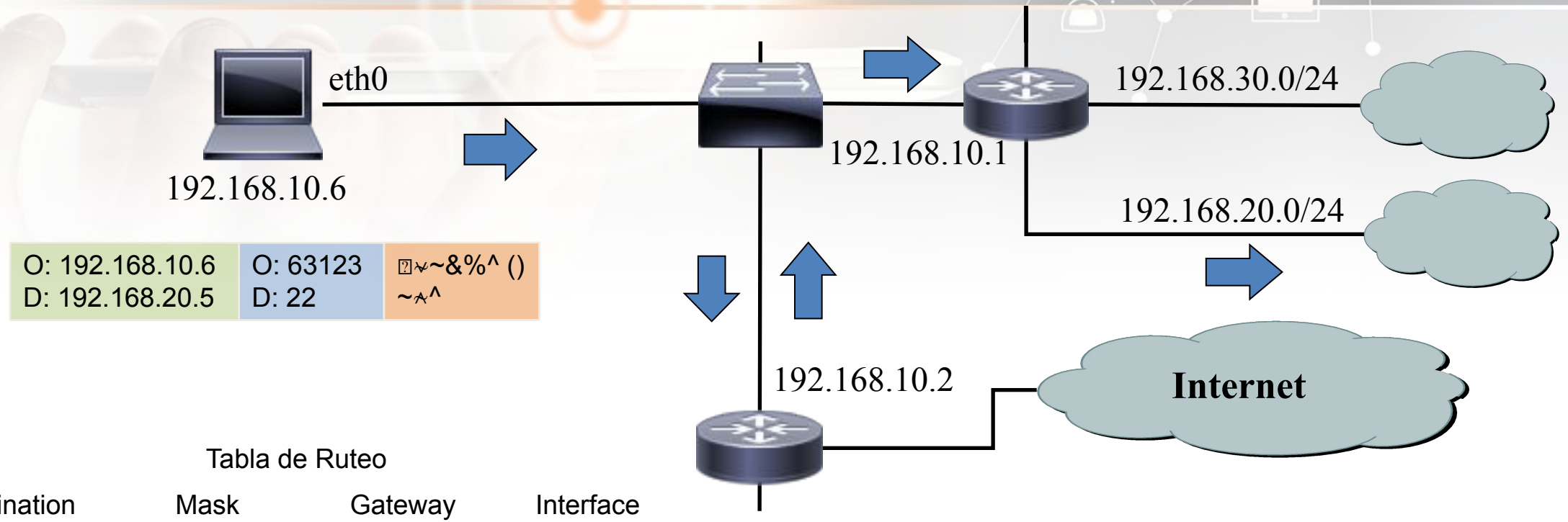
The checksum is the 16-bit ones's complement of the one's complement sum of the ICMP message starting with the ICMP Type. For computing the checksum , the checksum field should be zero. This checksum may be replaced in the future.

Gateway Internet Address

Internet Header + 64 bits of Data Datagram

The internet header plus the first 64 bits of the original datagram's data. This data is used by the host to match the message to the appropriate process. If a higher level protocol uses port numbers, they are assumed to be in the first 64 data bits of the original datagram's data.

ICMP : Redirect Message

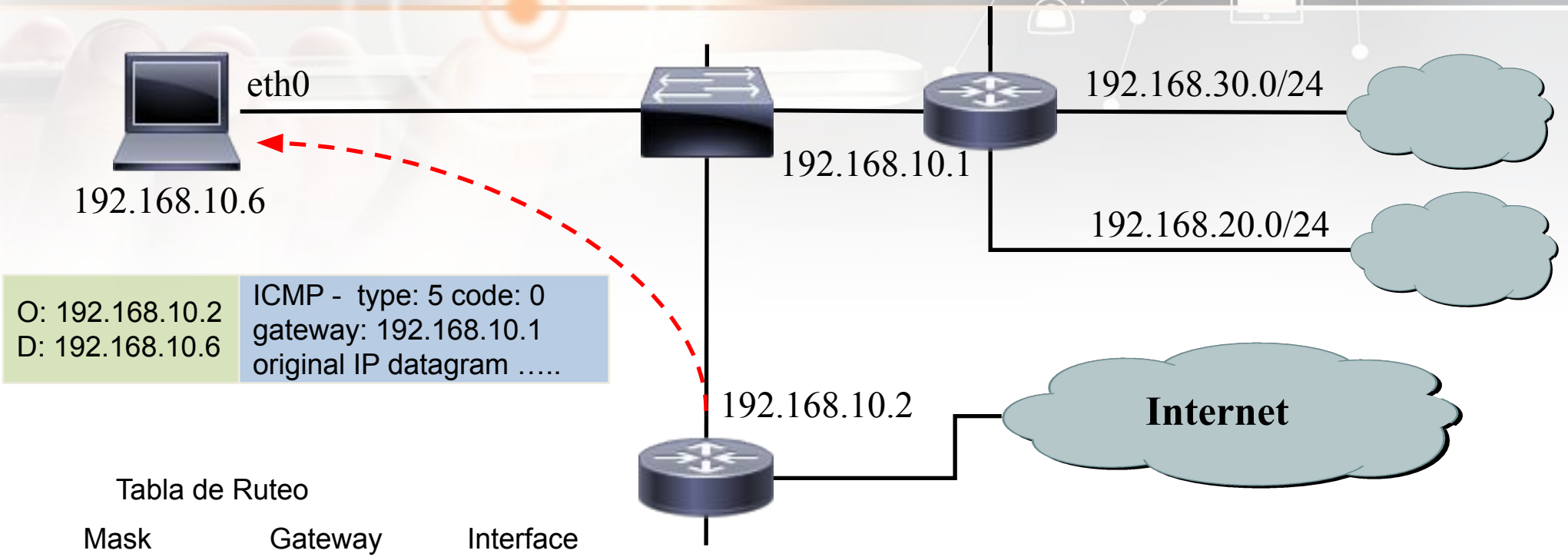


O: 192.168.10.6 D: 192.168.20.5	O: 63123 D: 22	⍰⍲~&%^ () ~⍝^

Tabla de Ruteo

Destination	Mask	Gateway	Interface
0.0.0.0	0.0.0.0	192.168.10.2	eth0
192.168.10.0	255.255.255.0	0.0.0.0	eth0
127.0.0.0	255.0.0.0	0.0.0.0	lo
169.254.0.0	255.255.0.0	0.0.0.0	eth0

ICMP : Redirect Message



O: 192.168.10.2 D: 192.168.10.6	ICMP - type: 5 code: 0 gateway: 192.168.10.1 original IP datagram …..

Tabla de Ruteo

Destination	Mask	Gateway	Interface
0.0.0.0	0.0.0.0	192.168.10.2	eth0
192.168.10.0	255.255.255.0	0.0.0.0	eth0
192.168.20.0	**255.255.255.0**	**192.168.10.1**	**eth0**
127.0.0.0	255.0.0.0	0.0.0.0	lo
169.254.0.0	255.255.0.0	0.0.0.0	eth0

ICMP : Echo Request / Reply Message

```
                                                          September 1981
RFC 792


Echo or Echo Reply Message

 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|     Type      |     Code      |          Checksum             |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|           Identifier          |        Sequence Number        |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|     Data ...
+-+-+-+-+-
```

```
Data (65500 bytes)
    Data: 6162636465666768696a6b6c6d6e6f707172737475767761…
    [Length: 65500]

0000  08 00 70 ce 00 01 02 ae  61 62 63 64 65 66 67 68   ··p····· abcdefgh
0010  69 6a 6b 6c 6d 6e 6f 70  71 72 73 74 75 76 77 61   ijklmnop qrstuvwa
0020  62 63 64 65 66 67 68 69  6a 6b 6c 6d 6e 6f 70 71   bcdefghi jklmnopq
0030  72 73 74 75 76 77 61 62  63 64 65 66 67 68 69 6a   rstuvwab cdefghij
0040  6b 6c 6d 6e 6f 70 71 72  73 74 75 76 77 61 62 63   klmnopqr stuvwabc
0050  64 65 66 67 68 69 6a 6b  6c 6d 6e 6f 70 71 72 73   defghijk lmnopqrs
```

```
Type

   8 for echo message;

   0 for echo reply message.

Code

   0

Checksum

   The checksum is the 16-bit ones's complement of the one's
   complement sum of the ICMP message starting with the ICMP Type.
   For computing the checksum , the checksum field should be zero.
   If the total length is odd, the received data is padded with one
   octet of zeros for computing the checksum.  This checksum may be
   replaced in the future.

Identifier

   If code = 0, an identifier to aid in matching echos and replies,
   may be zero.

Sequence Number

   If code = 0, a sequence number to aid in matching echos and
   replies, may be zero.

Description

   The data received in the echo message must be returned in the echo
   reply message.

   The identifier and sequence number may be used by the echo sender
   to aid in matching the replies with the echo requests.  For
   example, the identifier might be used like a port in TCP or UDP to
   identify a session, and the sequence number might be incremented
   on each echo request sent.  The echoer returns these same values
   in the echo reply.

   Code 0 may be received from a gateway or a host.
```

ICMP : Echo Request / Reply Message

```
user@server:~$ ping 200.49.213.50

PING 200.49.213.50 (200.49.213.50) 56(84) bytes of data.
64 bytes from 200.49.213.50: icmp_seq=1 ttl=55 time=21.4 ms
64 bytes from 200.49.213.50: icmp_seq=2 ttl=55 time=37.2 ms
64 bytes from 200.49.213.50: icmp_seq=3 ttl=55 time=35.7 ms
64 bytes from 200.49.213.50: icmp_seq=4 ttl=55 time=25.2 ms
64 bytes from 200.49.213.50: icmp_seq=5 ttl=55 time=23.2 ms

--- 200.49.213.50 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4036ms
rtt min/avg/max/mdev = 21.424/28.601/37.273/6.594 ms
```

ICMP : Echo Request / Reply Message

```
user@server:~$ ping -c1 200.49.213.50 -r
PING 200.49.213.50 (200.49.213.50) 56(124) bytes of data.
64 bytes from 200.49.213.50: icmp_seq=1 ttl=56 time=2286 ms
RR:    192.168.2.10
    200.26.75.69
    200.51.241.170                    200.51.241.57
    172.30.248.46
    209.13.133.250
    209.13.133.57
    200.51.240.82
    200.51.241.161
--- 200.49.213.50 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 2286.637/2286.637/2286.637/0.000 ms
```

Utilidades básicas de networking

En caso de tener problemas de conectividad, el comando ping nos puede ayudar a localizar dónde se encuentra el problema.

Comando	¿Qué verifica?
ping 127.0.0.1	que la pila de networking esté funcionando y que los servicios de red estén habilitados
ping dirección_IP_local	que la placa de red esté configurada.
ping host_en_mi_segmento	que tanto la placa como el cable estén funcionando correctamente. Así también como el switch que los vincula y que el host remoto responde.
ping host_en_otro_segmento	que el router o la tabla de ruteo (dependiendo de la complejidad de la red) esté configurado correctamente
ping host_fuera_de_la_red	que los routers con salida hacia Internet o redes externas funcionen en conjunto con los routers de la organización si los hubiera

Tener en cuenta que muchos routers o hosts no responden solicitudes de eco

Traceroute

```
user@server:~$ traceroute campus.itba.edu.ar
Tracing route to campus.itba.edu.ar [192.230.225.58]
  1     <1 ms     <1 ms     <1 ms  192.168.1.1
  2      *         *         *     Request timed out.
  ...
  6     15 ms     15 ms     15 ms  129-161-89-200.fibertel.com.ar [200.89.161.129]
  7     13 ms     15 ms     16 ms  130-165-89-200.fibertel.com.ar [200.89.165.130]
  8     13 ms     14 ms     11 ms  222-165-89-200.fibertel.com.ar [200.89.165.222]
  9      *         *         *     Request timed out.
 10    139 ms    140 ms    138 ms  ae1-300G.ar5.MIA1.gblx.net [67.17.94.249]
 11      *         *         *     Request timed out.
 12      *         *         *     Request timed out.
 13    160 ms    163 ms    159 ms  BLACKBOARD.ear2.Washington1.Level3.net
[4.59.144.218]
 14    165 ms    159 ms    164 ms  eth4-1-bb1-dc2.blackboard.com [69.196.255.133]
 15    159 ms    169 ms    160 ms  eth4-3-bb1-va2.mhint [69.196.255.145]
 16    166 ms    174 ms    175 ms  eth3-3-gw1-va2.mhint [69.196.255.158]
 17    158 ms    159 ms    159 ms  itba.blackboard.com [192.230.225.58]
```

IPv6

IP v4

Direcciones de 32 bits
4.294.967.296

Formato de dirección
192.168.1.1 / 24

Encabezado variable
Permite fragmentación
Estructura jerárquica (1994)
Configuración por DHCP o manual
Las direcciones se deben reutilizar y enmascarar
IPSec opcional

RFC 791 (septiembre 1981)

IP v6

Direcciones de 128 bits
340.282.366.920.938.463.463.374.607.431.768.211.456

Formato de dirección
fe80:0000:0000:94db:946e:8d4e:129e / 64

Encabezado fijo
No permite fragmentación
Estructura jerárquica de direcciones
Permite la configuración automática
Todos los dispositivos pueden tener una dirección exclusiva
IPSec incorporado en el diseño

RFC 2460 (diciembre 1998)

Notación Abreviada de direcciones IPv6

2001:0db8:0000:0000:0000:ff00:0042:8329

2001:db8:0:0:0:ff00:42:8329

2001:db8::ff00:42:8329

2001:0000:0000:0db8:0000:0000:0042:8329

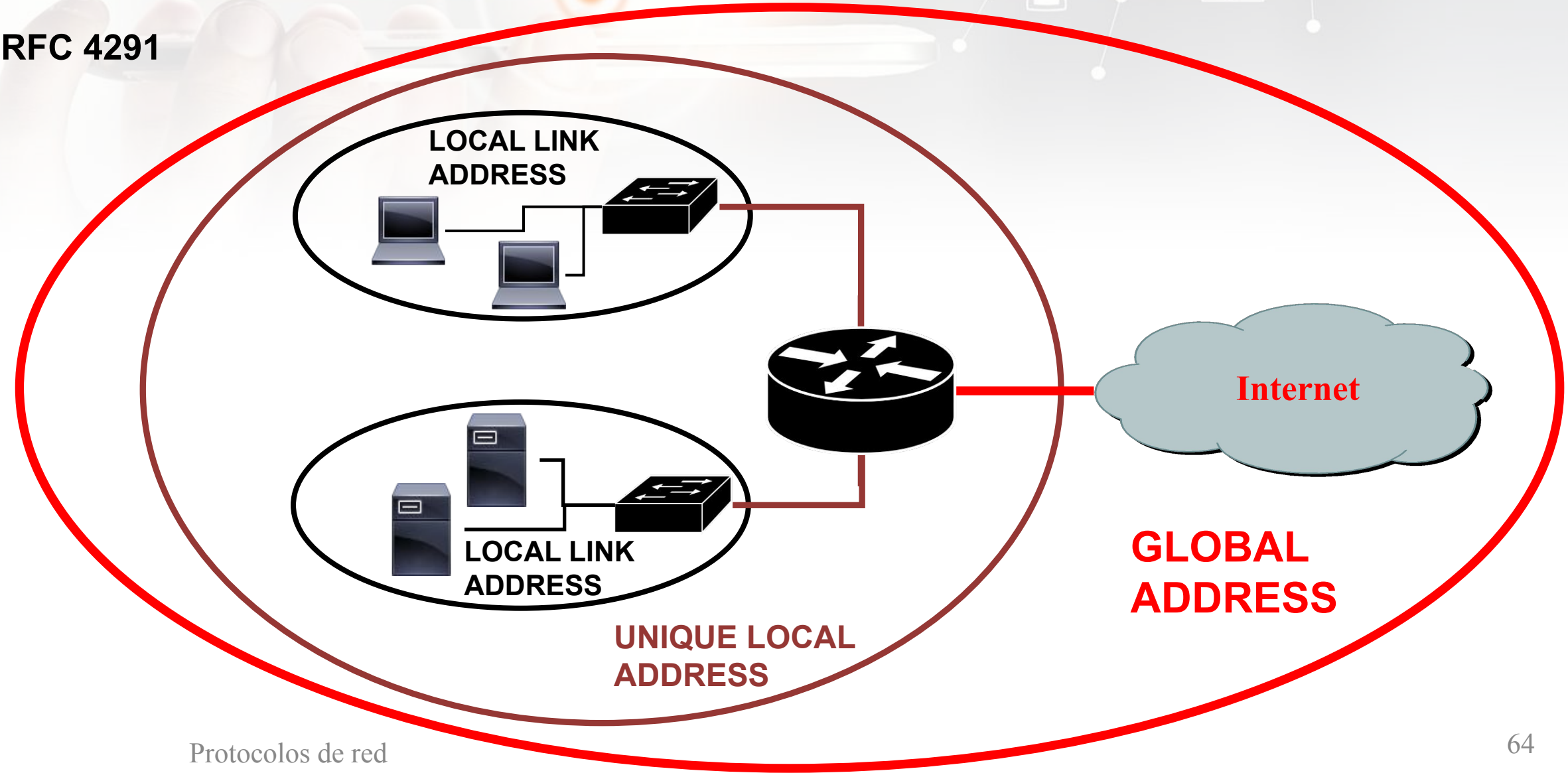
2001::db8:0:0:42:8329

2001:0:0:db8::42:8329

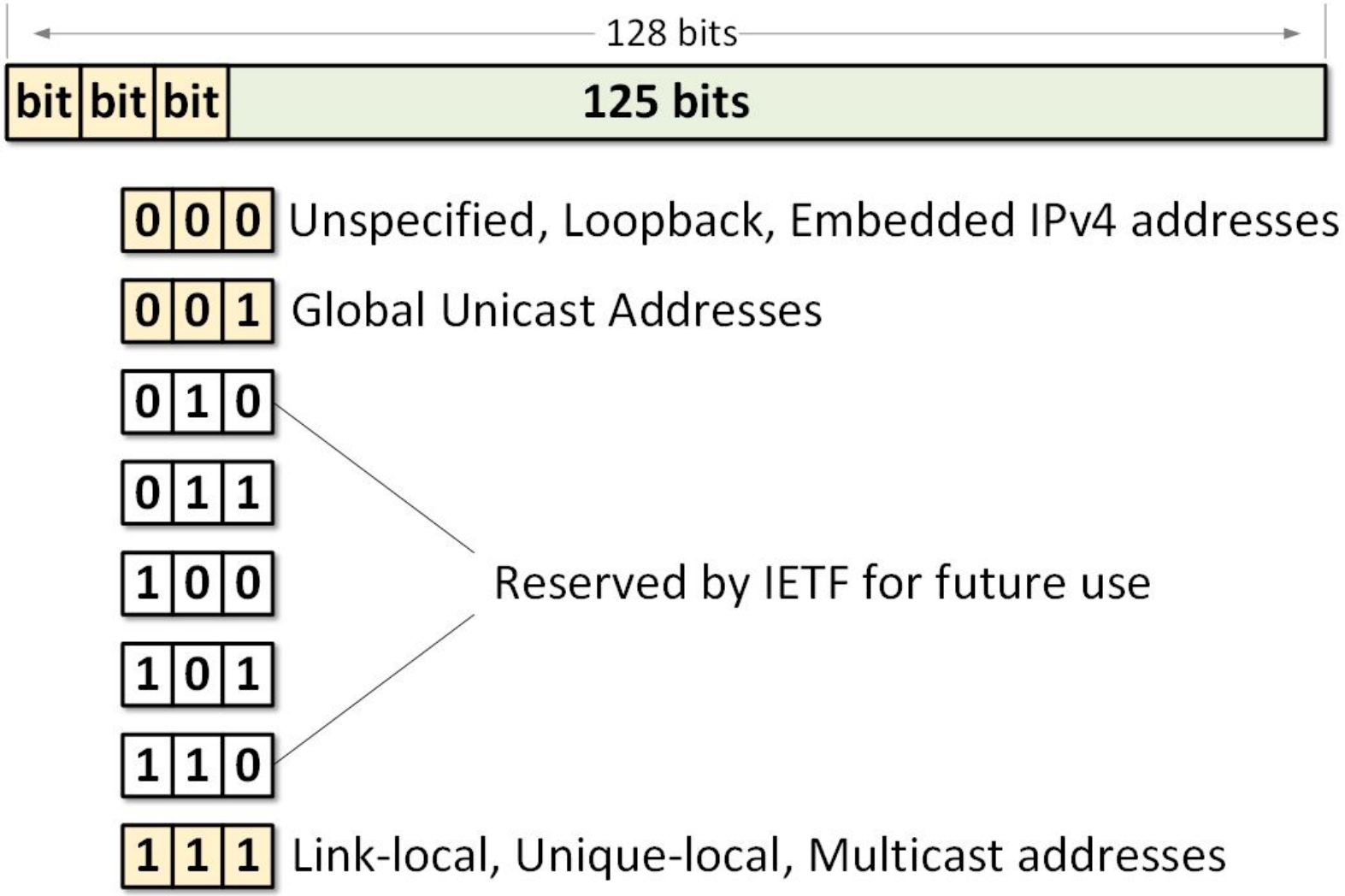
~~2001::db8::42:8329~~

Tipos de direcciones IPv6

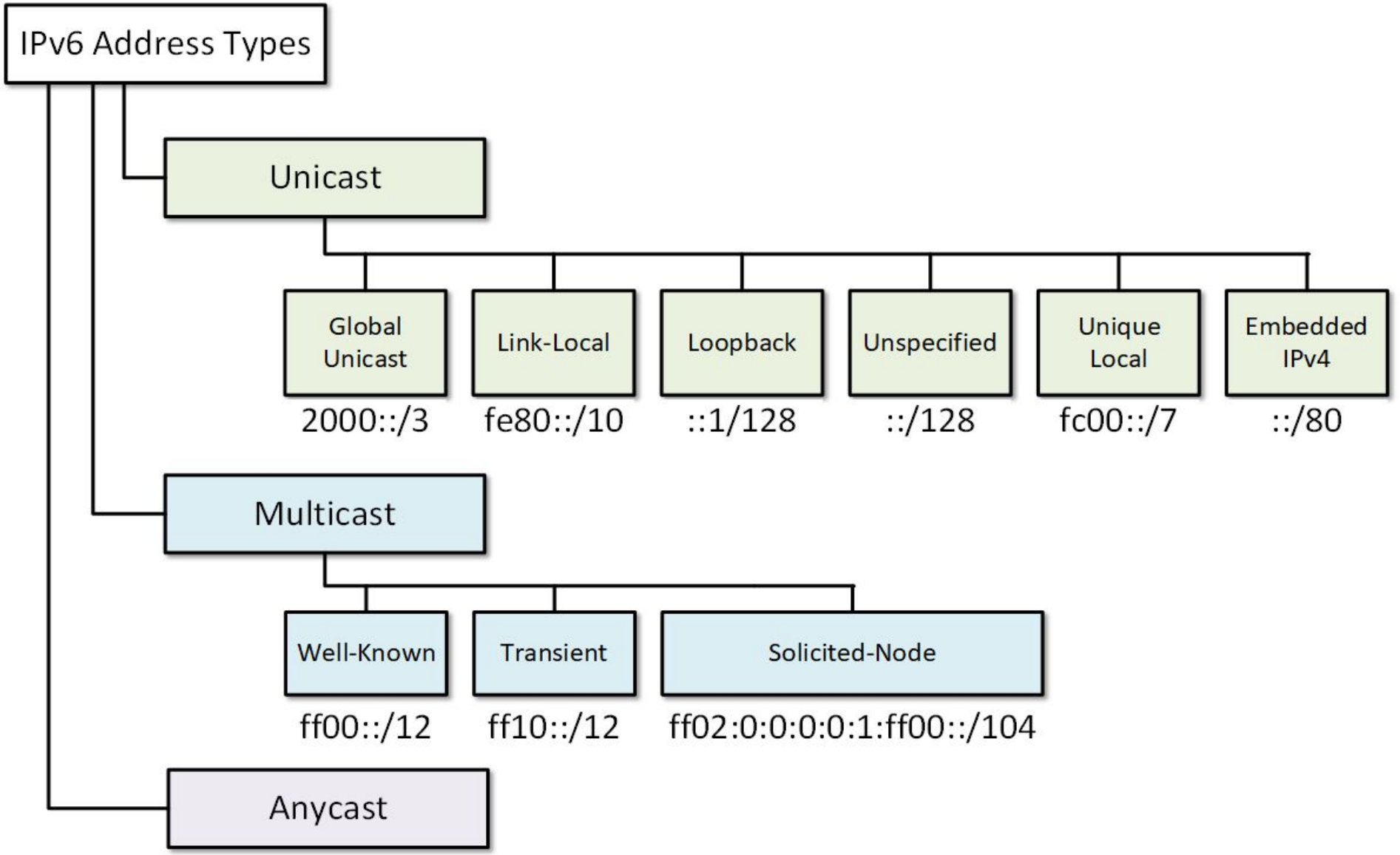
RFC 4291



Tipos de direcciones IPv6



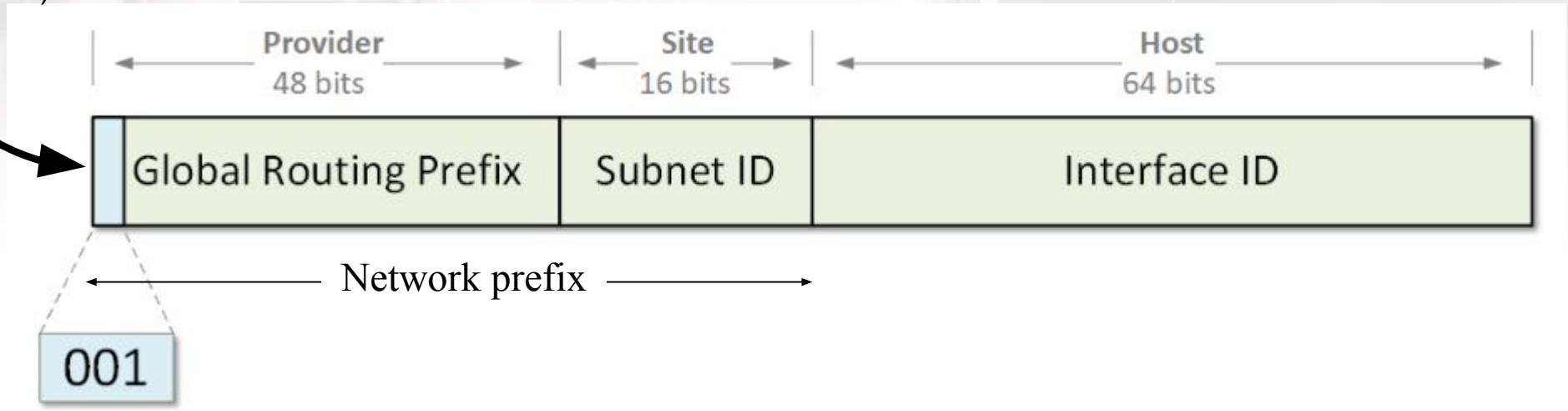
Tipos de direcciones IPv6



Formato de IPv6 Global Unicast Address

RFC 4291 (2006)

Prefijo 2000::/3



RFC 2374 (1998)

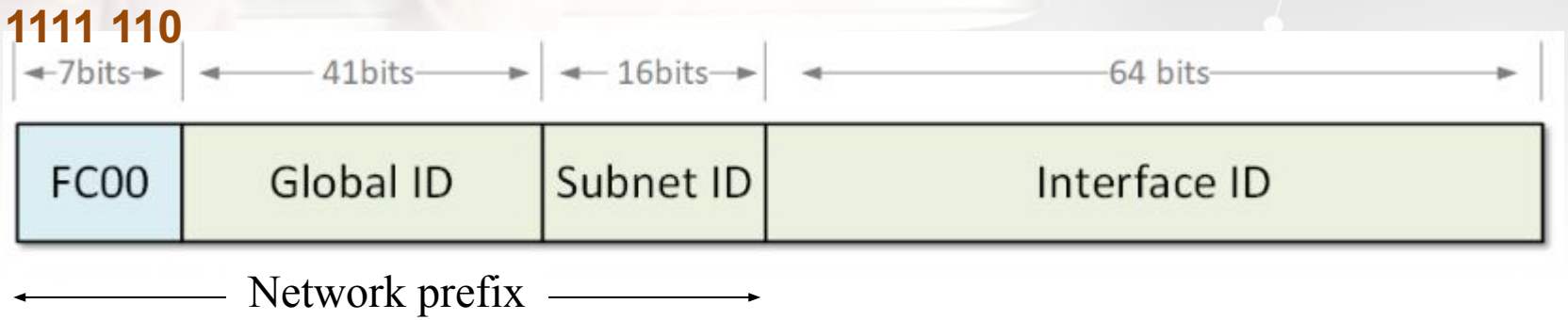
- Divide los 45 bits disponibles del prefijo de ruteo en dos bloques.
  - TLA ID (13 bits)
  - NLA ID (32 bits)

RFC 3587 (2003)

- Permite que los RIR gestionen el espacio de direcciones (45 bits de prefijo de ruteo) asignados por la IANA a su discreción.

Formato de IPv6 Unique Local Address

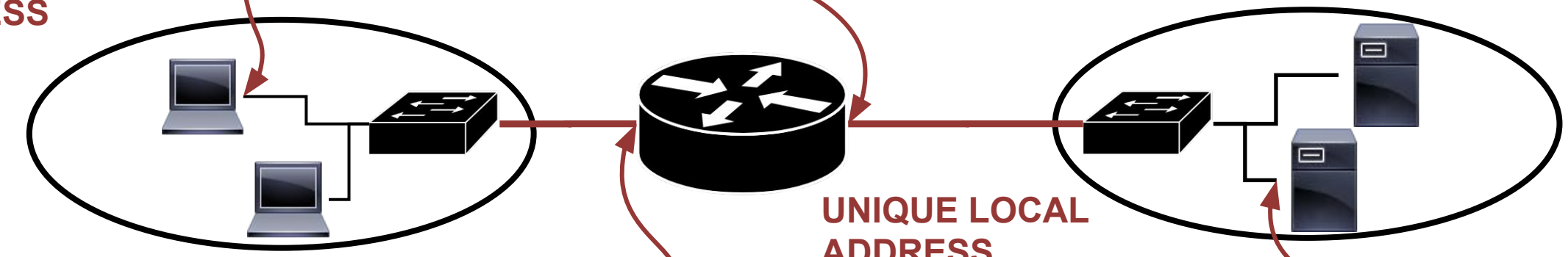
Prefijo FC00::/7



FC1A:2B3C:4D5E:6F00:0001:ABCD:EF01:2345

UNIQUE LOCAL ADDRESS

FC1A:2B3C:4D5E:6F11:0001:ABCD:0FE0:0001



UNIQUE LOCAL ADDRESS

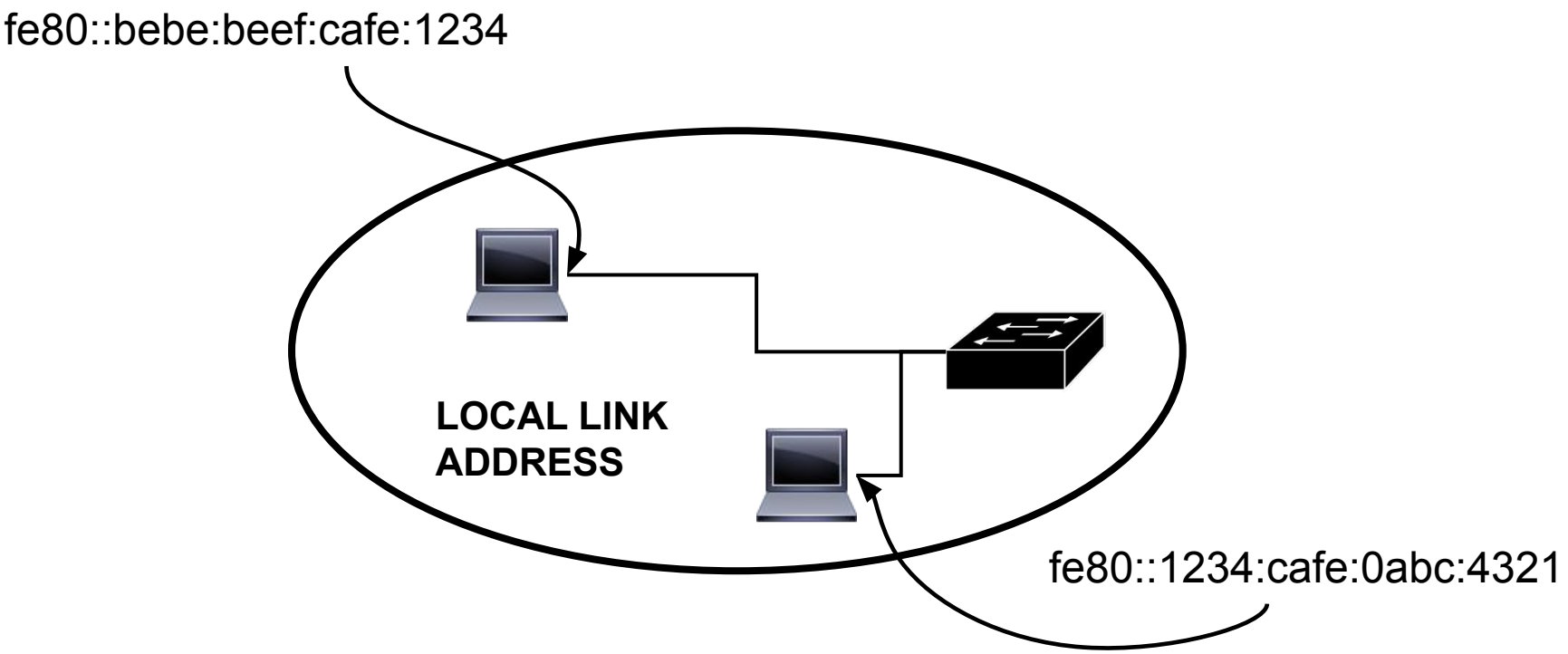
FC1A:2B3C:4D5E:6F00:0001:ABCD:EF01:0001

FC1A:2B3C:4D5E:6F11:0001:ABCD:0FE0:2345

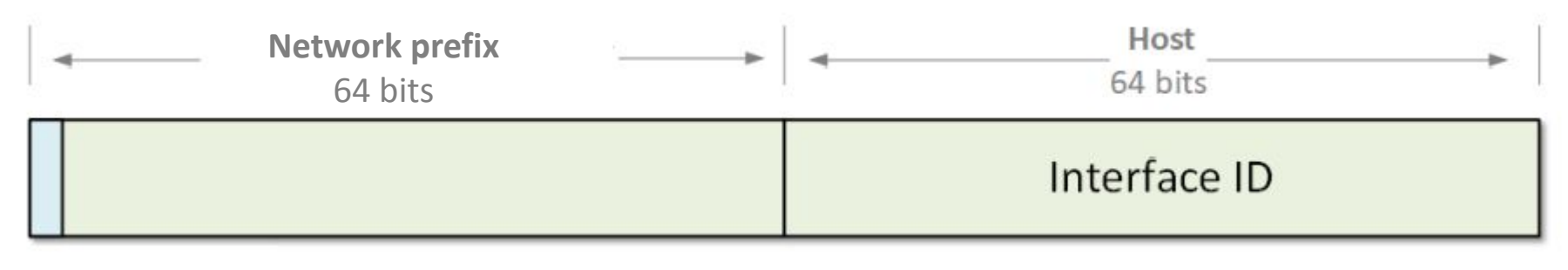
Formato de IPv6 Local Link Address

Prefijo FE80::/10

64 bits (16 hex digits)	64 bits (16 hex digits)
1111 1110 1000 0000 0000 0000...	Interface ID
10 bits / 54 bits	Typically derived from the MAC address



Asignación del Interface ID



EUI – 64

- Se basa en la dirección MAC. Ej 02:1A:2B:3C:4D:5E
- FE80:0000:0000:**021a:2b***FF:FE***3c:4d5e**
- En desuso por razones de privacidad y seguridad.

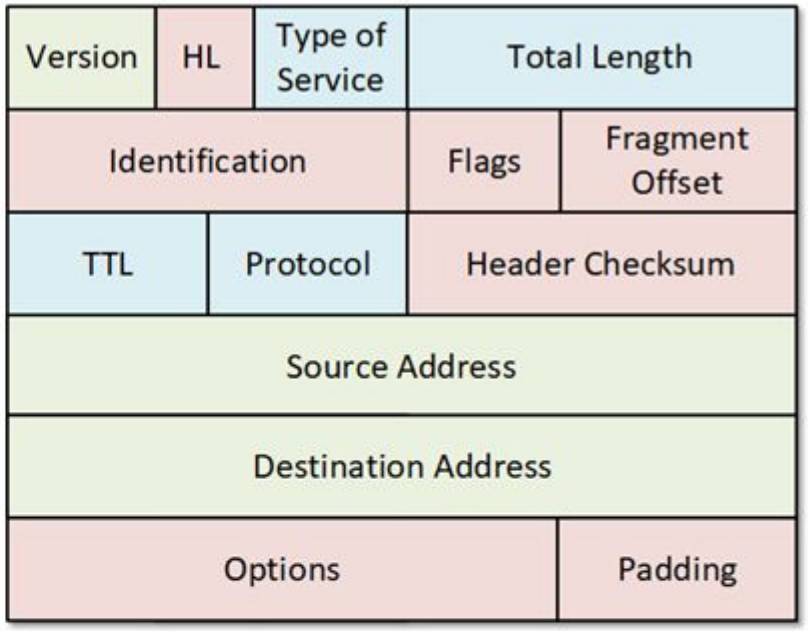
RFC 7217

- Genera direcciones de forma aleatoria, pero estables.
- En red A:
  - FE80:0000:0000:a1b2:c3d4:e5f6:7890
- El mismo host en red B:
  - FE80:0000:0000:bebe:beef:cafe:1234

Encabezado IPv6

RFC 8200 (2017)

Header IPv6



Header IPv4

Legend

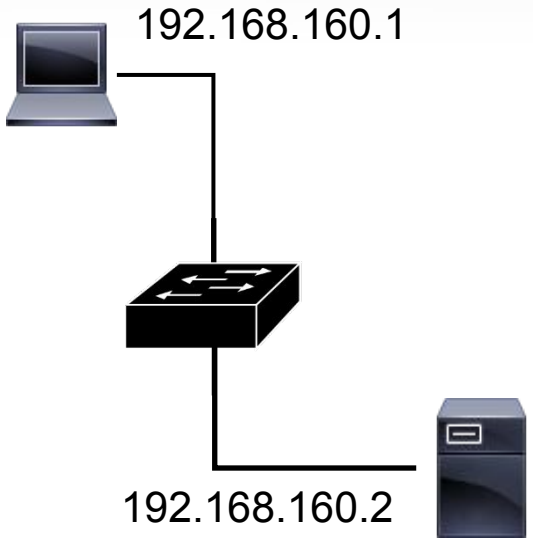
- Fields **kept** in IPv6
- Fields **kept** in IPv6, but name and position changed
- Fields **not kept** in IPv6
- Fields that are **new** in IPv6

IPv4-Mapped IPv6

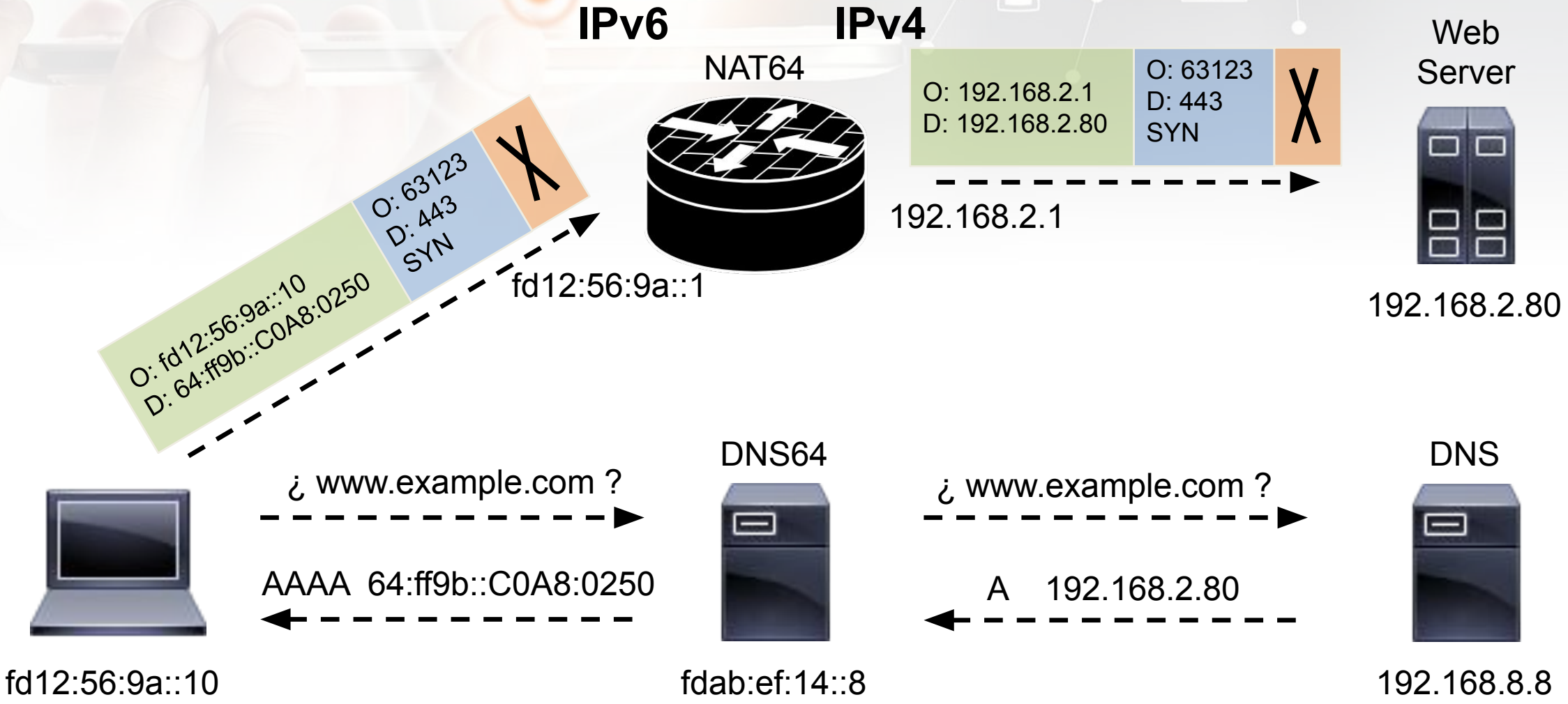
RFC 4291

```
|                 80 bits                | 16 |      32 bits        |
+----------------------------------------+----+---------------------+
|0000.............................0000|FFFF|    IPv4 address     |
+----------------------------------------+----+---------------------+
```

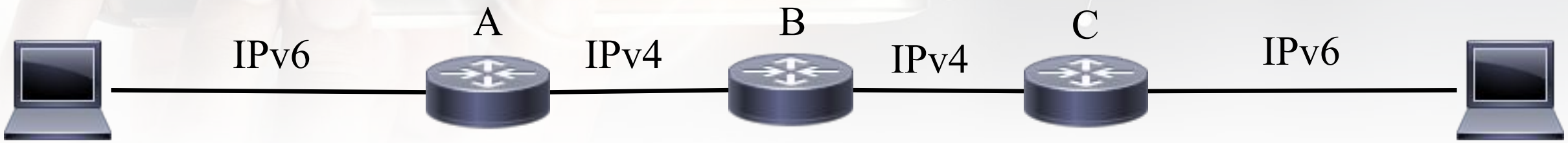
```
C:\Users>ping ::FFFF:192.168.160.2

Haciendo ping a 192.168.160.2 con 32 bytes de datos:
Respuesta desde 192.168.160.2: bytes=32 tiempo=1ms TTL=255
Respuesta desde 192.168.160.2: bytes=32 tiempo=1ms TTL=255
Respuesta desde 192.168.160.2: bytes=32 tiempo=5ms TTL=255
Respuesta desde 192.168.160.2: bytes=32 tiempo=2ms TTL=255

Estadísticas de ping para 192.168.160.2:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
    (0% perdidos),
Tiempos aproximados de ida y vuelta en milisegundos:
    Mínimo = 1ms, Máximo = 5ms, Media = 2ms

C:\Users>ping ::FFFF:C0A8:A002

Haciendo ping a 192.168.160.2 con 32 bytes de datos:
Respuesta desde 192.168.160.2: bytes=32 tiempo=1ms TTL=255
Respuesta desde 192.168.160.2: bytes=32 tiempo=1ms TTL=255
Respuesta desde 192.168.160.2: bytes=32 tiempo=1ms TTL=255
Respuesta desde 192.168.160.2: bytes=32 tiempo=2ms TTL=255

Estadísticas de ping para 192.168.160.2:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
    (0% perdidos),
Tiempos aproximados de ida y vuelta en milisegundos:
    Mínimo = 1ms, Máximo = 2ms, Media = 1ms
```

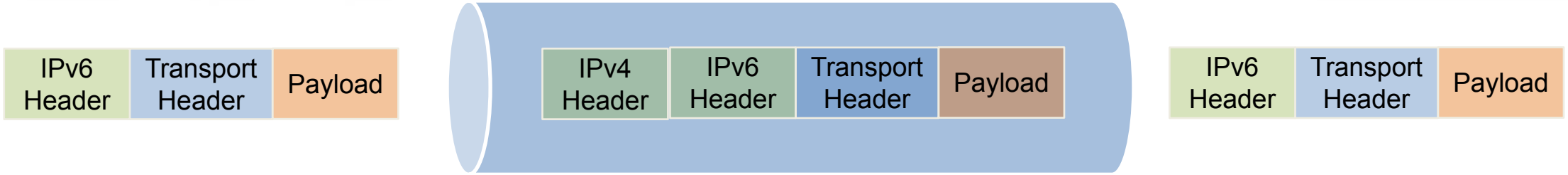


NAT64: conectando IPv6 con IPv4



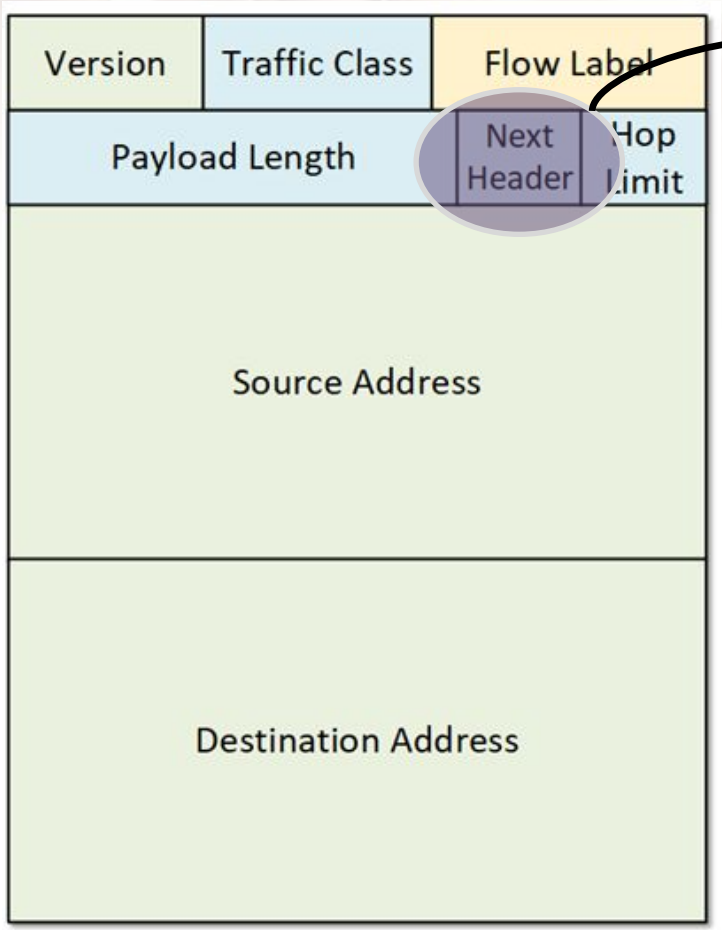
Tunneling



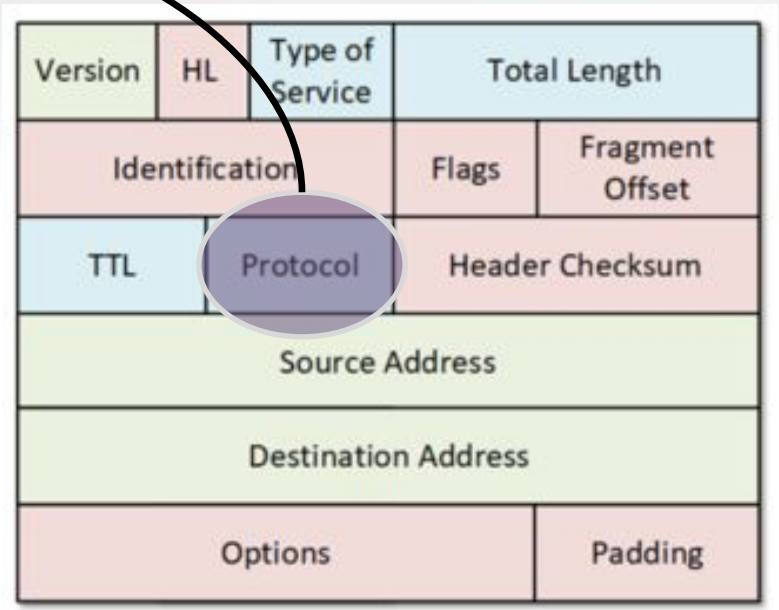


Tunneling: consiste en encapsular datos de un protocolo en los datos de otro protocolo de **igual o mayor** nivel.

Tipos de Túneles

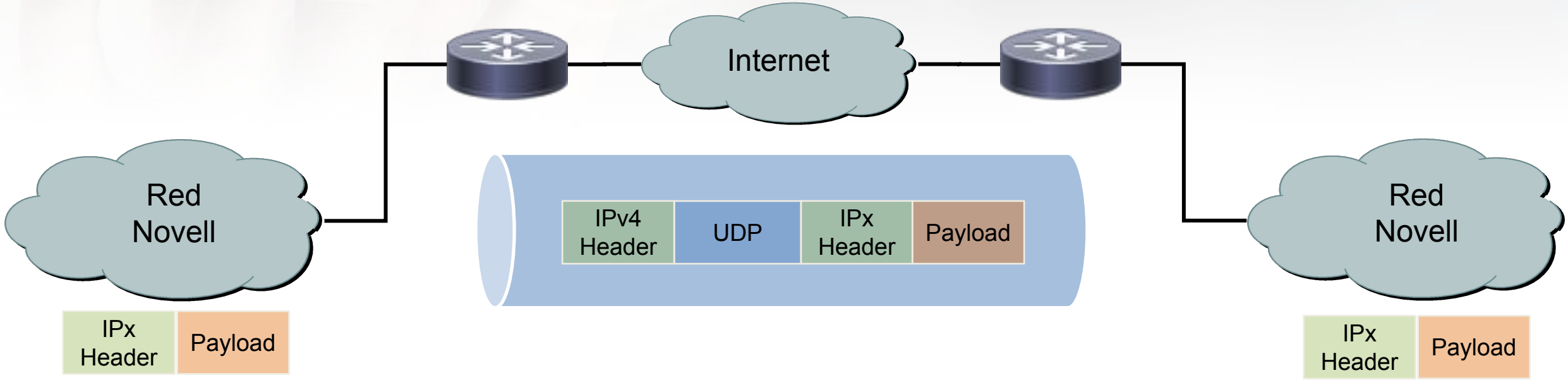


Valor	Prot.
1	ICMP
2	IGMP
4	IP
6	TCP
17	UDP
41	IPv6
89	OSPF



https://www.iana.org/assignments/protocol-numbers/protocol-numbers.xhtml

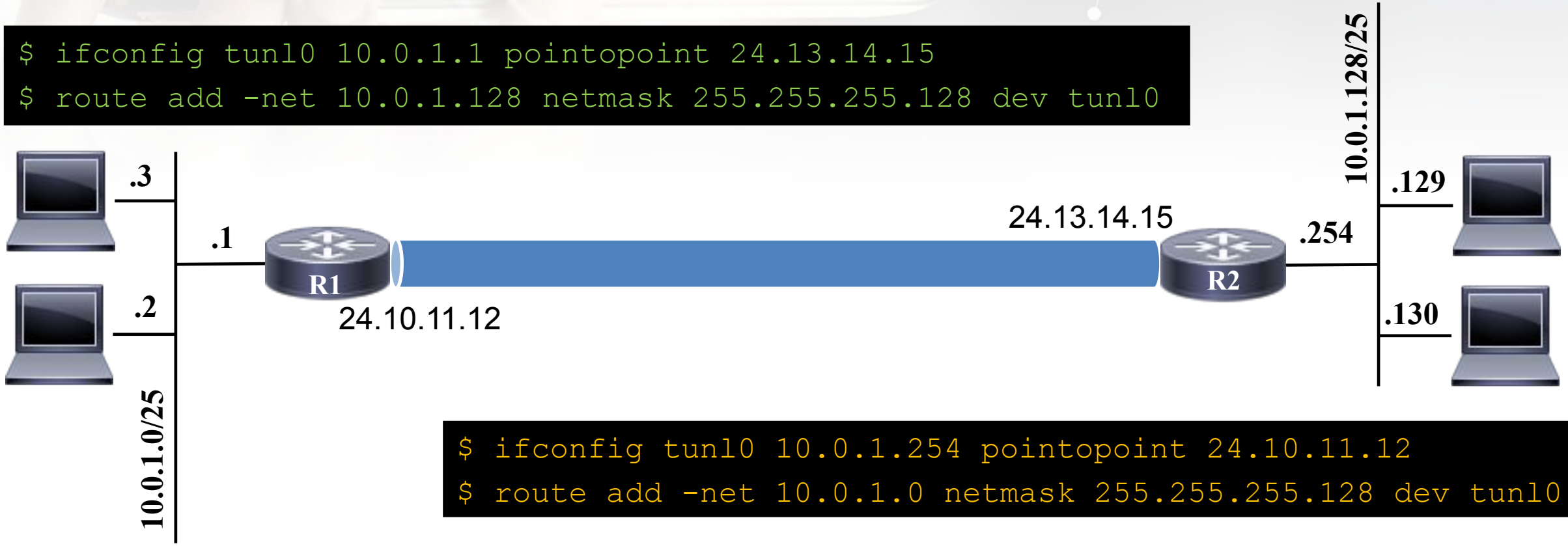
Túneles de Protocolos Desconocidos



Tunneling: consiste en encapsular datos de un protocolo en los datos de otro protocolo de **igual o mayor** nivel.

Túneles IP-en-IP

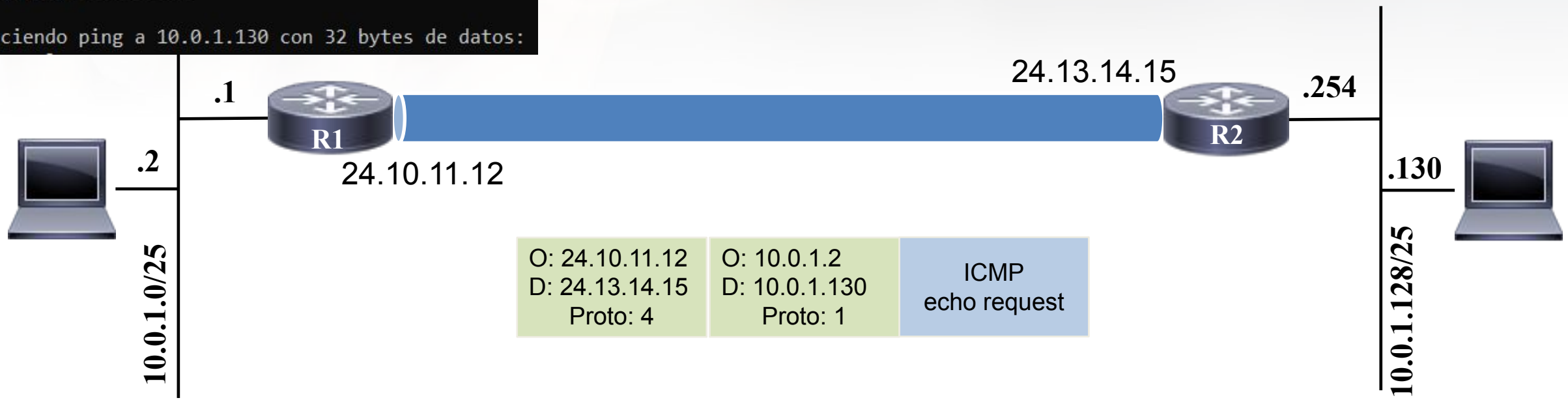
```
$ ifconfig tunl0 10.0.1.1 pointopoint 24.13.14.15
$ route add -net 10.0.1.128 netmask 255.255.255.128 dev tunl0
```




```
$ ifconfig tunl0 10.0.1.254 pointopoint 24.10.11.12
$ route add -net 10.0.1.0 netmask 255.255.255.128 dev tunl0
```

Túneles IP-en-IP

```
C:\>ping 10.0.1.130

Haciendo ping a 10.0.1.130 con 32 bytes de datos:
```



O: 24.10.11.12 D: 24.13.14.15 Proto: 4	O: 10.0.1.2 D: 10.0.1.130 Proto: 1	ICMP echo request

O: 10.0.1.2 D: 10.0.1.130 Proto: 1	ICMP echo request

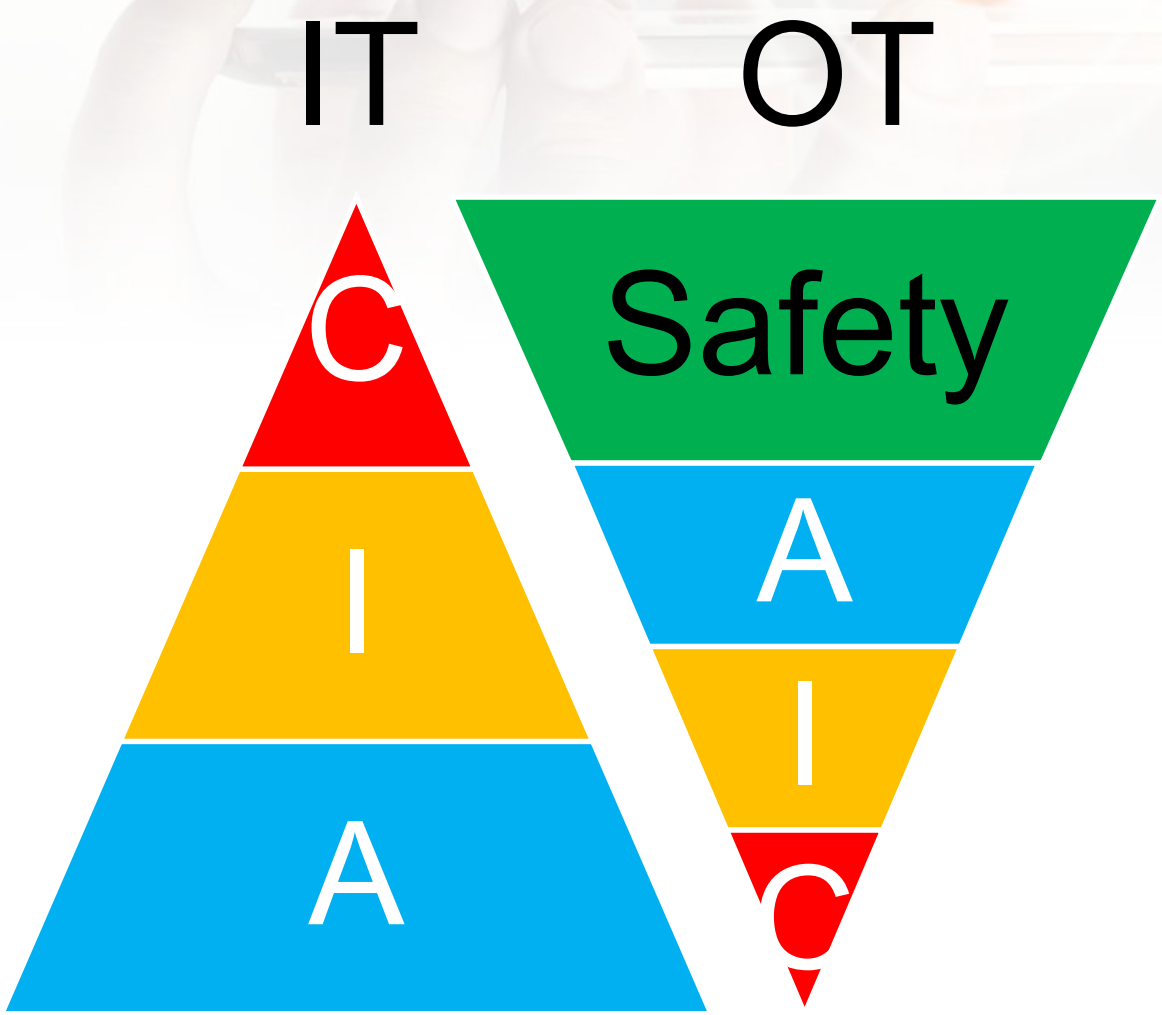
¿ Confidencialidad y Integridad ?

O: 10.0.1.2 D: 10.0.1.130 Proto: 1	ICMP echo request

Material de lectura

Capítulos 4.1 a 4.4 inclusive de la bibliografía

Seguridad



C – Confidencialidad
Garantiza que la información solo sea accesible a personas autorizadas. Se enfoca en proteger los datos contra accesos no autorizados. Ejemplos:

- *Cifrado de datos*
- *Control de acceso / Autenticación*

I – Integridad
Asegura que la información no sea alterada de manera no autorizada. Se trata de mantener los datos precisos y completos. Ejemplos:

- *Hashing*
- *Firmas digitales*

A – Disponibilidad
Garantiza que la información y los sistemas estén accesibles cuando se necesiten. Se enfoca en evitar interrupciones del servicio. Ejemplos:

- *Redundancia de sistemas*
- *Copias de seguridad*
- *Protección contra ataques DDoS*

IPSec

- Objetivo:

  Proveer servicios de seguridad para los paquetes IP en el nivel de capa de red.

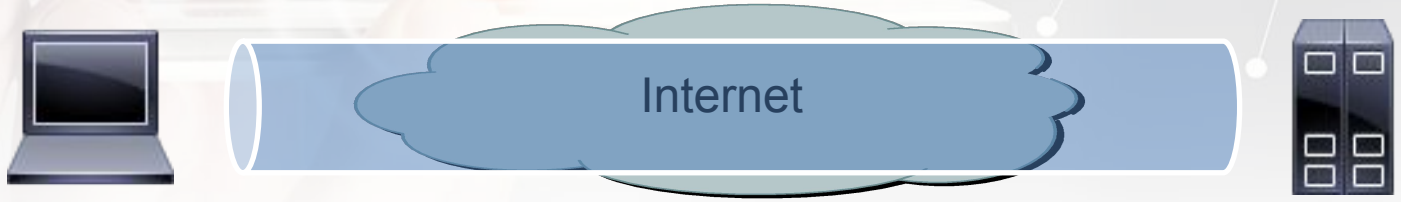
- Servicios:

  Control de Acceso
  Confidencialidad
  Autenticación
  Integridad

- Modos:

  Transporte
  Túnel

IPSec – Modo Transporte



IP proto : 6	TCP	Payload

IP proto : 6	TCP	Payload

Authentication Header →

IP proto : 51	AH	TCP	Payload

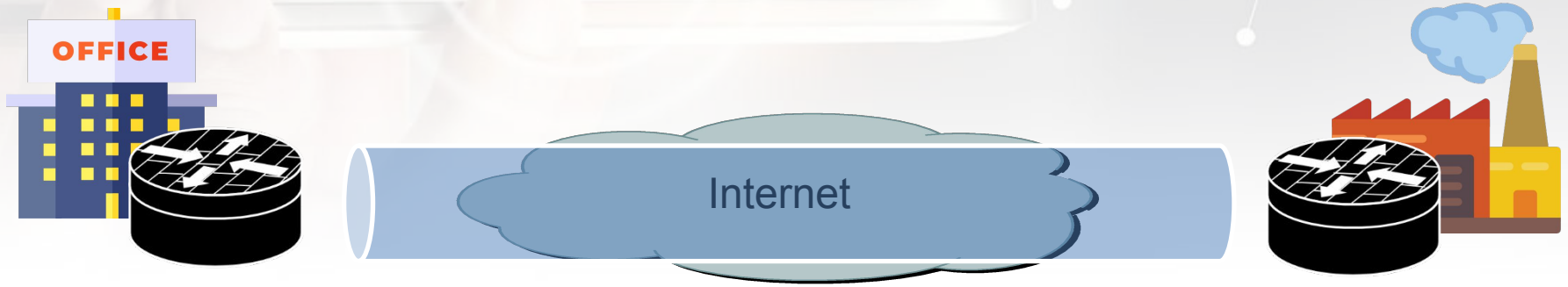
- ✅ Protege contra ataques de modificación de paquetes. Integridad y Autenticidad.
- ❌ No protege la confidencialidad, los datos viajan en texto claro.

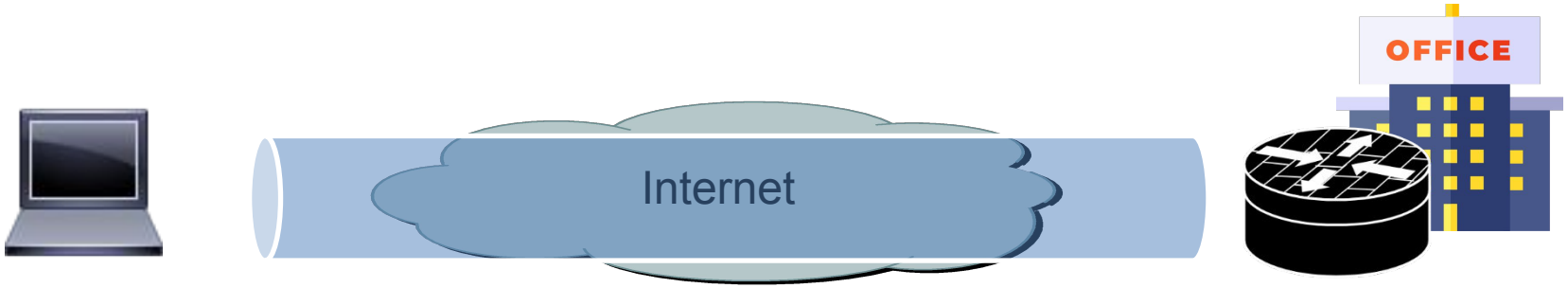
Encapsulating Security Payload →

IP proto : 50	ESP Header	TCP	Payload	ESP Trailer	ESP Auth

- ✅ Confidencialidad, Integridad y Autenticación

IPSec – Modo Túnel





IPSec – Modo Túnel – AH y ESP

IP proto : 6	TCP	Payload

Authentication Header →

IP proto : 51	AH	IP proto : 6	TCP	Payload

✅ Protege **integridad y autenticidad**, incluyendo la cabecera IP original.
❌ **No cifra los datos**, solo los autentica.

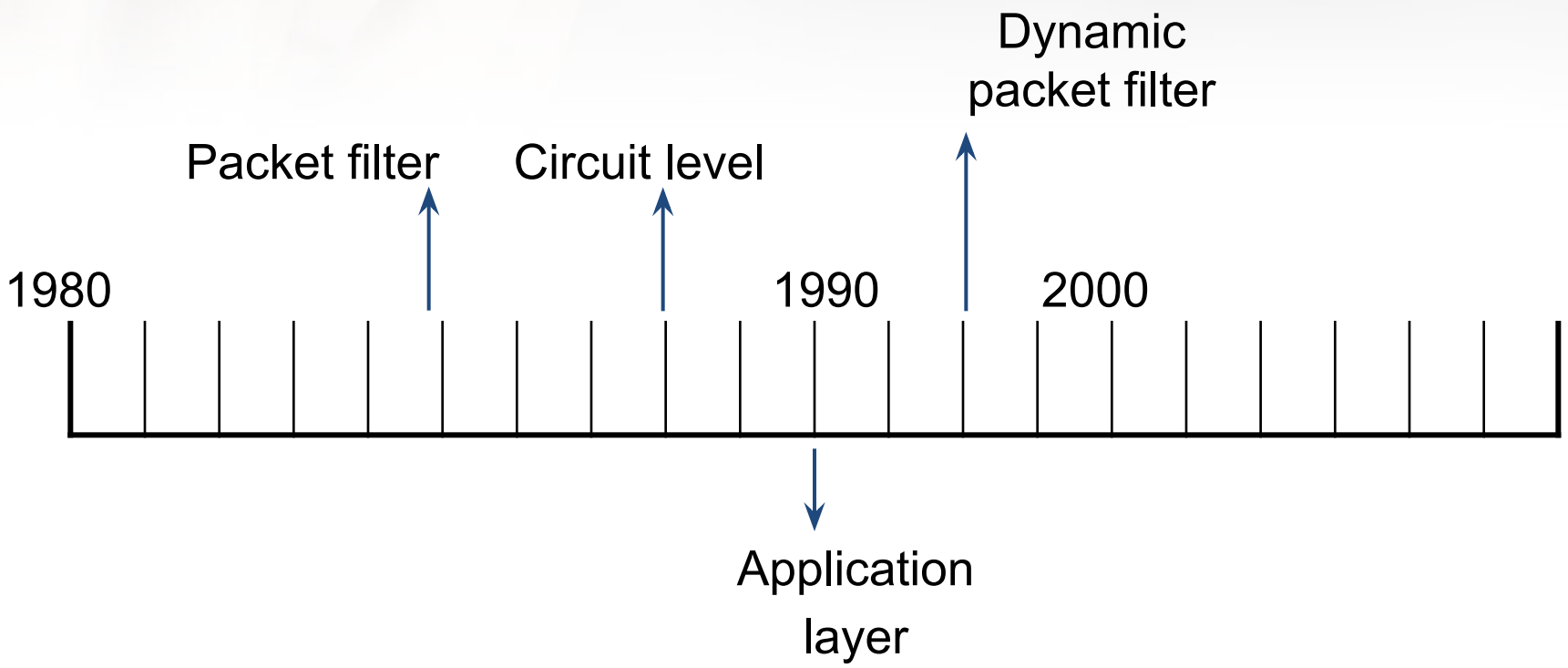
Encapsulating Security Payload →

IP proto : 50	ESP Header	IP proto : 6	TCP	Payload	ESP Trailer	ESP Auth

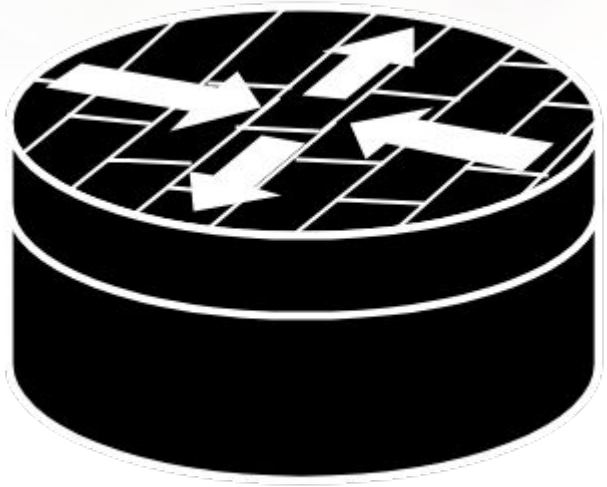
✅ **Cifra toda la comunicación** (cabecera IP original y datos).
✅ **Protege confidencialidad, integridad y autenticidad**.

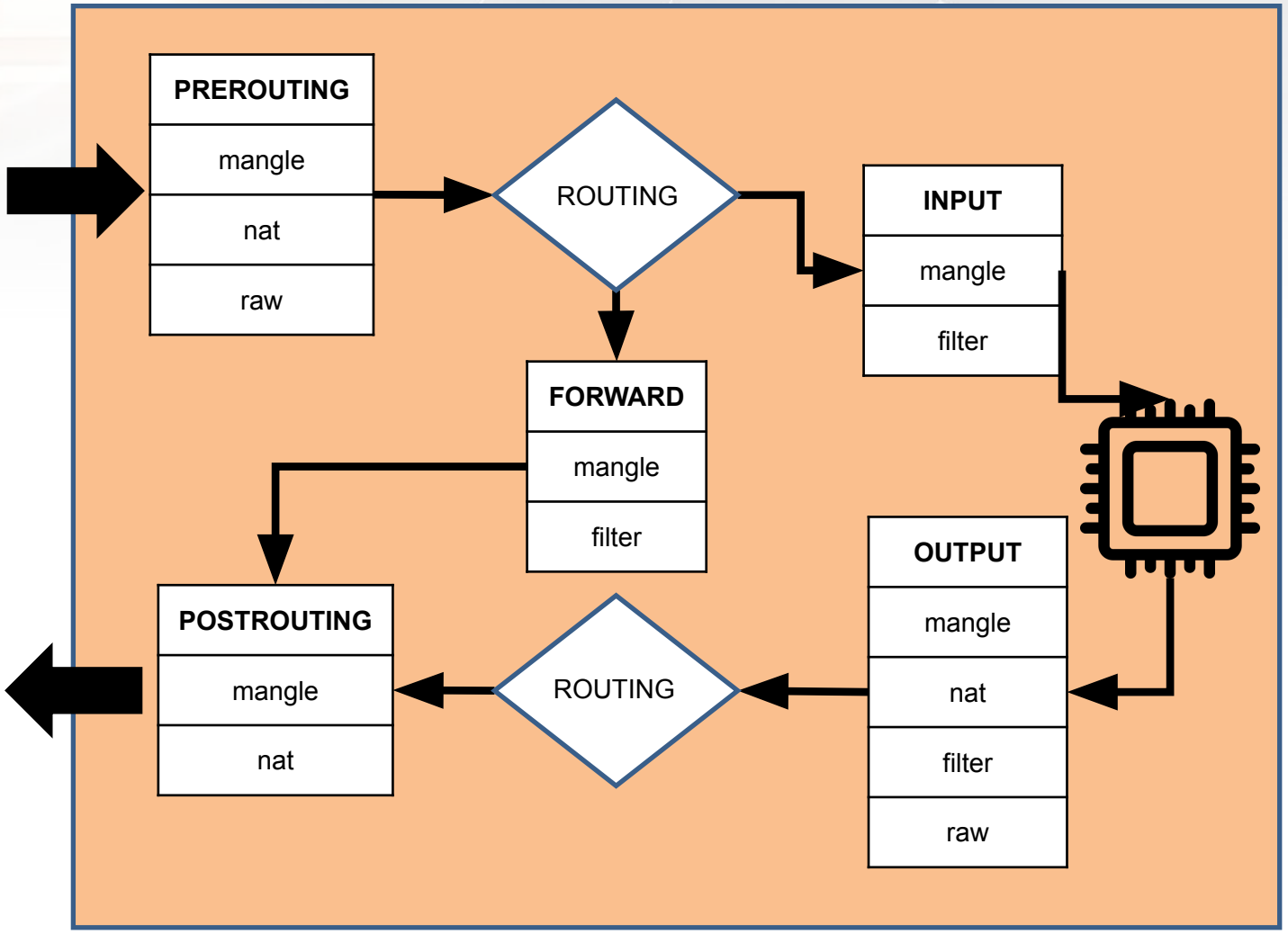
Firewalls

Evolución de la arquitectura de firewalls



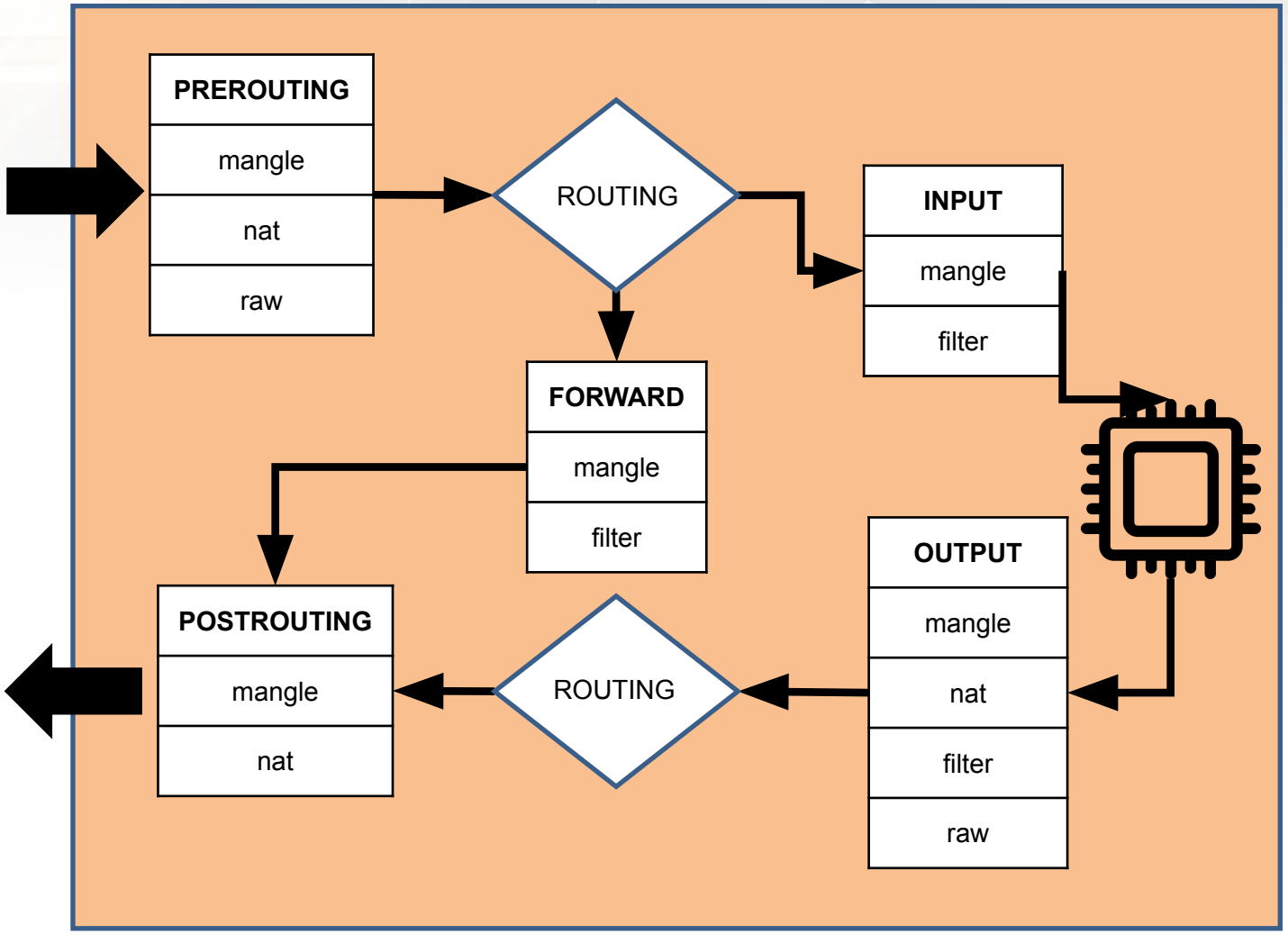
IPTables




PREROUTING
mangle
nat
raw
ROUTING
INPUT
mangle
filter
FORWARD
mangle
filter
OUTPUT
mangle
nat
filter
raw
POSTROUTING
mangle
nat
ROUTING

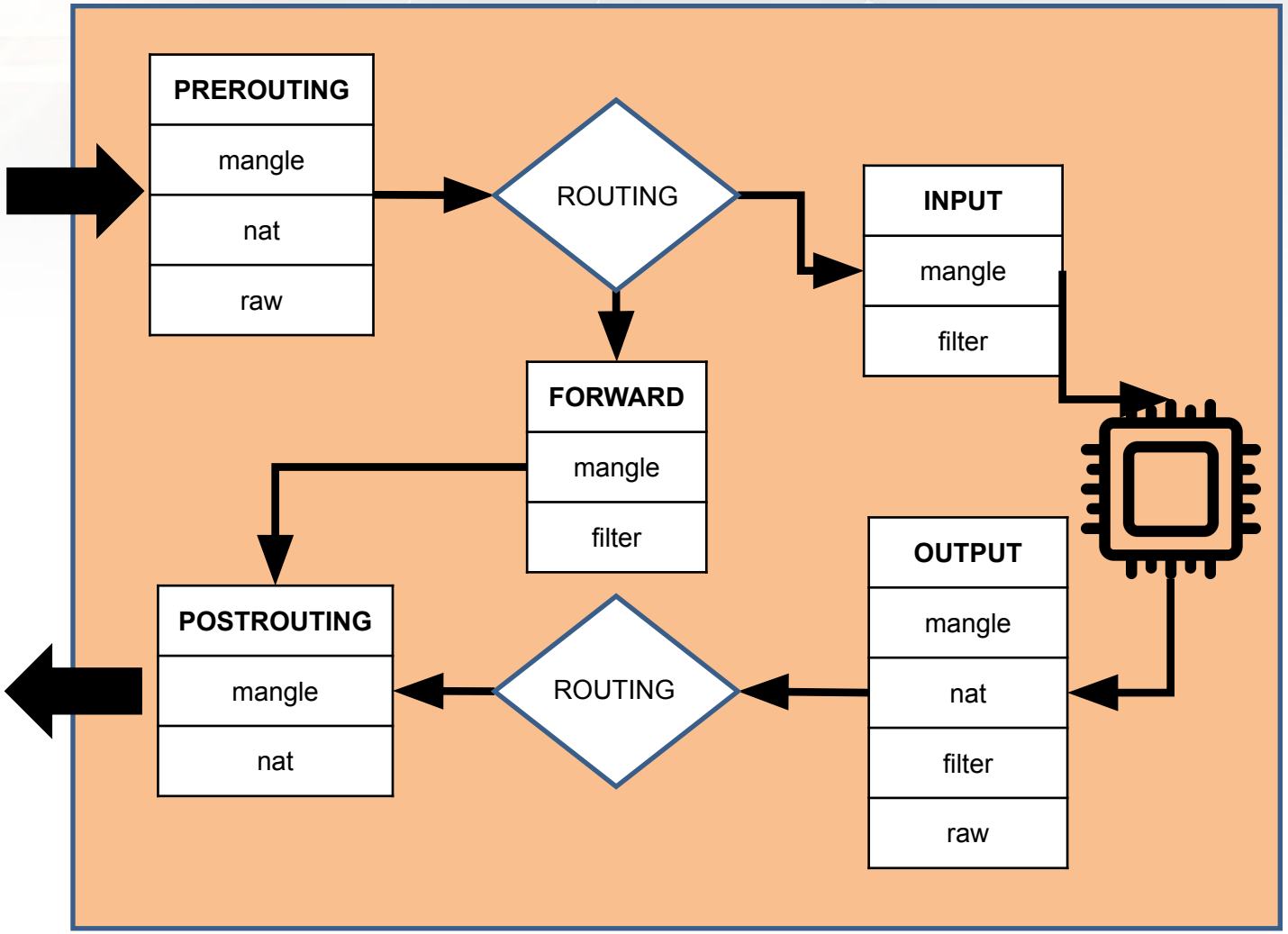
IPTables

Cadenas	¿Dónde actúa?
PREROUTING	Antes del enrutamiento, modifica paquetes antes de rutear.
INPUT	Controla paquetes que entran al sistema y están dirigidos a él.
FOWARD	Maneja tráfico que atraviesa el sistema
OUTPUT	Controla los paquetes generados por el sistema antes de salir.
POSTROUTING	Modifica paquetes después de que se ha ruteado.



IPTables

Tablas	¿Dónde actúa?
filter	Filtrado de paquetes (permitir, denegar, registrar).
nat	Traducción de direcciones (DNAT, SNAT Masquerade).
mangle	Modificación avanzada de paquetes (ToS, TTL, marcado).
raw	Manipulación de paquetes antes del seguimiento de conexiones.



IPTables – Comandos Básicos

Listar reglas de la tabla NAT en IPTables

```
~$ iptables -t nat -L

Chain PREROUTING (policy ACCEPT)
target     prot opt source       destination

Chain POSTROUTING (policy ACCEPT)
target     prot opt source       destination
MASQUERADE all  --  192.168.1.0/24  anywhere

Chain OUTPUT (policy ACCEPT)
target     prot opt source       destination
```

IPTables – Comandos Básicos

Listar conexiones NAT activas

```
~$ netstat-nat -n

Proto NATed Address          Foreign Address     State
tcp   192.168.1.100:443  ->  20.28.101.14:443  ESTABLISHED
tcp   192.168.1.50:22    ->  20.28.101.14:22   TIME_WAIT
udp   192.168.1.200:53   ->  20.28.101.14:53   ASSURED
```

IPTables – Comandos Básicos

Tres formas de aplicar SNAT, dado que la interfaz eth1 es la WAN

```
~$ iptables -t nat -A POSTROUTING -o eth1 -j MASQUERADE

~$ iptables -t nat -A POSTROUTING -s 192.168.1.0/24 -o eth1 -j SNAT --to 20.28.101.14

~$ iptables -t nat -A POSTROUTING -p tcp -o eth1 -j SNAT --to 20.28.101.14:1-1023
```

IPTables – Ejemplo Práctico

```
~$ cat config.sh
#!/bin/bash
# Reenviar todos los paquetes de eth1 (red interna) a eth0 (Internet).
iptables -A FORWARD -i eth1 -o eth0 -j ACCEPT

# Habilitar SNAT  para IP pública fija:
iptables –t nat -A POSTROUTING -o eth0 –j SNAT –to-source 24.232.138.143
```

IPTables – Ejemplo Práctico

```
# No aceptar IP Spoofing
iptables -A INPUT -i eth0 -s 24.232.138.143/32 -j DROP
iptables -A INPUT -i eth0 -s 192.168.0.0/24 -j DROP
iptables -A INPUT -i eth0 -s loopback/8 -j DROP

# Aceptamos lo que venga de localhost
iptables -A INPUT -i lo -j ACCEPT
```

IPTables – Ejemplo Práctico

```
# Implementamos DNAT para un Web Server en un host local
iptables –t nat –A PREROUTING –d 24.232.138.143 -p tcp –dport 80 –j DNAT /
 --to-destination 192.168.2.3
```